

GRADUATE CERTIFICATE

INTELLIGENT REASONING SYSTEM (IRS)

---------- **PROJECT REPORT** ----------

Banking Voice Agent for Complaint Resolution and Responsible Next Best Action Recommendations

---

Table of Contents

1 Introduction..... 4
1.1 Background and Problem Statement..... 4
1.2 Project Aim and Objectives..... 5
1.3 Project Overview and Scope..... 5
2 Project Background..... 6
2.1 Singapore Retail Banking Service Context..... 6
2.2 Service Complexity and Operational Dependencies..... 6
2.3 Evidence and Uncertainty in Banking Assistance..... 7
2.4 Accountability and Privacy Requirements..... 7
3 Literature Review and Market Research..... 8
3.1 Market Landscape: Conversational AI in Retail Banking..... 8
3.2 Technical Review: Intent Recognition and Retrieval-Augmented Generation..... 8
3.3 Hybrid Reasoning Architectures and Auditable AI Governance..... 9
4 System Architecture and Model..... 9
4.1 System Overview..... 9
4.2 Retrieval-Augmented Generation Model..... 11
4.3 Recommendation Decision Model..... 11
4.4 Intent, Risk and Entity Understanding..... 12
4.5 Rules and Complaint State Machine..... 14
4.6 LLM Assisted Agent Safety Boundary..... 15
4.7 Optional Voice Input Architecture and Safety Boundary..... 16
5 System Development and Implementation..... 16
5.1 BANKING77 Dataset and Preparation..... 16
5.2 Retrieval-Augmented Generation..... 17
5.2.1 Knowledge Base Construction..... 17
5.2.2 FAQ Expression Mapping..... 18
5.2.3 BM25 Candidate Retrieval..... 18
5.2.4 Cross-Encoder Reranking..... 19
5.3 Product Recommendation..... 19
5.3.1 Synthetic Data and Validation..... 19
5.3.2 Eligibility and Deterministic Ranking..... 20
5.3.3 Defensive Outcomes and Integration..... 21
5.4 Intent, Risk and Entity Understanding..... 21
5.4.1 Data Preparation and Operational Label Design..... 21
5.4.2 Classifier Training with a Frozen Sentence Encoder..... 22
5.4.3 Runtime Inference, Risk and Entity Processing..... 22
5.5 Rules and Complaint State Machine Implementation..... 23
5.6 LLM Gateway and Guarded Delivery Implementation..... 24
5.7 Local ASR Implementation..... 24
5.8 Web Application and Local Persistence..... 25
6 Experiment Results and Evaluation..... 26

6.1 RAG Retrieval Evaluation..... 26
6.2 Recommendation Module Validation..... 28
6.3 BANKING77 Intent Classification Evaluation..... 29
6.4 End-to-End System Evaluation..... 31
6.4.1 Single-turn End-to-End Results..... 31
6.4.2 Multi-turn State and Safety Results..... 32
6.4.3 LLM Delivery Reliability..... 32
6.4.4 Separate Local ASR Smoke Evaluation..... 33
7 Findings and Discussion..... 34
8 Conclusion and Future Work..... 34
9 Reference..... 35
10 Appendix..... 38
10.1 Declaration..... 38
10.2 Mapped System Functionalities against Modular Courses..... 38
10.3 Installation and User Guide..... 38
10.4 Project Proposal:..... 40

1 Introduction

1.1 Background and Problem Statement

Retail banking customer service requires conversational flexibility alongside consistent control over procedural actions. A single customer dialogue may interleave diverse operational intents—such as product enquiries, fee disputes, and reports of unauthorized transactions. Each pathway demands different evidential support and operational guardrails, making reliable automation dependent on both natural language understanding and strict procedural compliance.

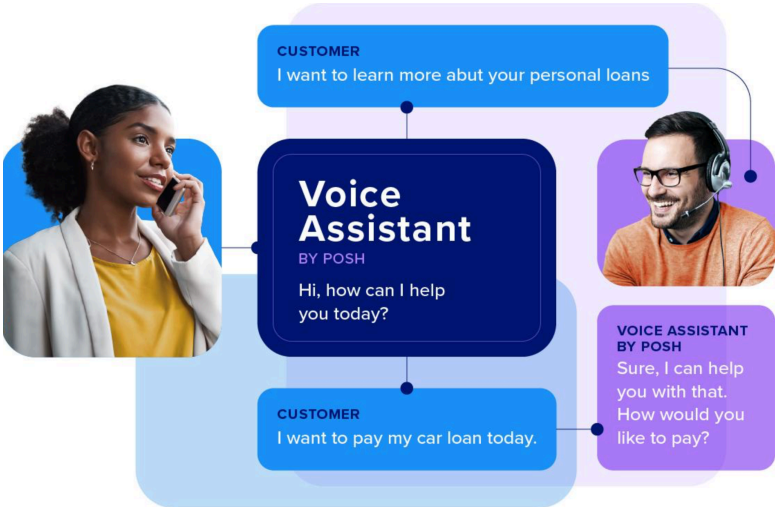


Figure 1: Conceptual illustration of an automated conversational banking workflow (adapted from Posh Technologies).

While spoken interaction introduces linguistic ambiguity, informal phrasing, and automatic speech recognition (ASR) noise, conventional rule-based engines lack conversational adaptability. Conversely, generative large language models (LLMs) offer fluent dialogue but risk generating ungrounded claims or actions misaligned with regulatory policies. The core engineering challenge is thus bridging flexible natural communication with verifiable, evidence-grounded, and procedurally governed decision-making.

To examine this tension, this project introduces **NovaBank**, a fictional Singapore retail banking environment. NovaBank combines natural language processing, policy retrieval, and deterministic state transitions to govern service workflows across enquiries, complaint lifecycles, and risk escalations.

1.2 Project Aim and Objectives

The project's central aim was to develop and evaluate a hybrid banking assistant for a fictional retail bank in Singapore. The resulting prototype combines intent and risk identification, evidence-supported answers, rule-governed complaint handling, and conditional next best action recommendations within an integrated workflow. Its design makes service decisions and supporting evidence available for inspection, enabling evaluation of both component outputs and workflow correctness.



The project has five objectives:

1. **Integrate the service workflow.** Connect customer input, intent recognition, risk assessment, retrieval, decision control, and response generation through consistent module interfaces.
2. **Interpret requests and uncertainty.** Identify service needs and risk signals, distinguishing ambiguous requests from questions that lack reliable supporting evidence.
3. **Provide grounded answers.** Retrieve relevant NovaBank policy information, retain source references, and prevent unsupported details from being presented as established policy.
4. **Control complaints and recommendations.** Track complaint progress, support appropriate escalation, and authorise the recommendation pathway only when the required service-resolution, consent, risk, confidence, and handoff conditions are satisfied. Apply product eligibility filtering and ranking after authorisation.
5. **Assess performance and traceability.** Evaluate component outputs and complete service pathways, using decision records to investigate both successful behaviour and failures.

1.3 Project Overview and Scope

NovaBank employs a modular hybrid architecture. Linguistic interpretation is handled by statistical classifiers, policy evidence is supplied through a two-stage retrieval pipeline (BM25 candidate selection**[1]** followed by Cross-Encoder reranking**[2]**), and service commitments are strictly gated by deterministic rules. A centralized orchestrator maintains conversation state and records decision traces across all routing checkpoints.The prototype targets routine banking services, fee disputes, suspected fraud escalations, and conditionally permitted product recommendations. To enable controlled investigation without operational risk:

- **Data & Environment**: Runtime demonstrations use synthetic customer profiles, fictional NovaBank policies and simulated handoffs. Public BANKING77 utterances are used separately for supervised intent-classification training and evaluation.
- **Interface Modalities:** Evaluation uses text input and output. Browser voice input now supports recording, editable ASR transcripts and confirmation before manual sending. TTS remains a future extension, with English-Mandarin support optional.

- **Operational Boundaries**: The assistant does not interface with live banking infrastructure, authenticate real customer identities, or execute binding financial transactions. Performance metrics validate the internal reasoning architecture rather than deployment readiness or statutory regulatory compliance.

2 Project Background

2.1 Singapore Retail Banking Service Context

Retail banking customer assistance operates at the intersection of natural conversational interaction and legally binding operational procedures. Service quality requires not merely conversational responsiveness, but strict adherence to institutional guidelines and appropriate channel routing.

Singapore's established dispute-resolution framework highlights this necessity for procedural continuity. Guidance from MoneySense and MAS expects financial institutions to resolve consumer grievances promptly and transparently, distinguishing initial receipt of a concern from substantive investigation and formal outcome notification **[3], [4]**. In an automated workflow, recording an issue serves merely as an intermediate operational event rather than case resolution. Consequently, conversational systems must maintain explicit state tracking to identify unresolved liabilities, govern permitted procedural transitions, and prevent reassuring yet factually deceptive responses.

2.2 Service Complexity and Operational Dependencies

Customer interactions in retail banking rarely follow static, pre-categorized paths. An initial card transaction enquiry can dynamically evolve into a billing dispute or reveal unauthorized fraudulent activity. Because security-related requests carry substantial risks, explicit risk signals should not be disregarded solely because intent-classification confidence is low. This principle concerns the treatment of uncertainty; the precedence between risk escalation and unsupported account-operation requests is specified separately.

Furthermore, escalation represents a structured transfer of operational context and accountability rather than a superficial channel handoff. When automated handling concludes, persistent state—including unresolved questions and completed verification steps—must be preserved for human personnel. Similarly, proactive product recommendations depend on operational context: unresolved complaints, lack of explicit consent, or elevated risk signals strictly override product eligibility.

2.3 Evidence and Uncertainty in Banking Assistance

Financial policy disclosures typically hinge on operational qualifications—such as verification thresholds, tiered tariff conditions, and dynamic transactional prerequisites—that conversational systems can easily omit or misinterpret. Ensuring factual and regulatory correctness requires far more than surface-level lexical matching or semantic similarity retrieval. The supporting policy corpus must explicitly satisfy the conditional predicates governing the customer's specific circumstances.

This operational imperative introduces a fundamental architectural decoupling between **linguistic uncertainty** and **evidential deficiency[5]**:

- **Linguistic Ambiguity (CLARIFY)**: Under-specified user utterances or missing required transaction entities necessitate iterative multi-turn clarification to establish the exact service request.
- **Evidential Deficiency (ABSTAIN)**: When a query is semantically unambiguous but the underlying knowledge base lacks authoritative or applicable policy documentation, requesting the customer to rephrase is epistemically flawed. In this scenario, the system must trigger an explicit abstention mechanism rather than synthesizing an ungrounded or hallucinated response.

By formalizing **CLARIFY** and **ABSTAIN** as mutually exclusive decision branches rather than treating all unresolved inputs as intent-classification failures, the system establishes an explicit rule-based fallback mechanism. Low classification confidence or insufficient policy evidence should not, by themselves, suppress a recognised security signal. The resulting service action remains subject to the decision order. This design assigns service-action authority to explicit rules rather than generated text. Its effectiveness depends on correct rule implementation and is assessed through workflow testing.

2.4 Accountability and Privacy Requirements

To support accountability and post-hoc inspection, NovaBank maintains structured decision traces linking module outcomes, policy-evidence references and business-state changes to individual requests. These records allow reviewers to examine the processing path and investigate unexpected decisions. They support traceability but do not, by themselves, establish tamper resistance or regulatory compliance.

This design directly operationalizes the Accountability and Transparency tenets set forth in the Monetary Authority of Singapore's (MAS) FEAT principles. Decision traces provide an auditable governance trail for verifying algorithmic behavior against procedural guidelines without requiring model parameter introspection.

Simultaneously, auditable logging must be balanced against statutory data-protection obligations under Singapore's Personal Data Protection Act (PDPA). Guidelines issued by the Personal Data Protection Commission (PDPC) mandate strict compliance regarding consent, purpose limitation, and retention limits for personally identifiable information (PII) embedded in transcripts and speech streams. System auditability cannot serve as justification for unrestrained data collection or unbounded log retention.**[6]**

Consequently, the NovaBank prototype was implemented within a fully isolated synthetic sandbox. By utilizing artificial customer personas, curated fictitious policy documentation, and simulated operational handoffs, this testbed enables controlled evaluation of the modular reasoning workflow, state transitions, and compliance boundaries without accessing live customer accounts or executing real financial transactions.

3 Literature Review and Market Research

3.1 Market Landscape: Conversational AI in Retail Banking

Automated customer assistance in modern retail banking bridges informational self-service and legally binding operational procedures. Leading financial institutions in Singapore increasingly deploy hybrid conversational systems: DBS incorporates generative AI within its *digibot* under explicit knowledge bounds, customer authentication, and safety guardrails; UOB tightly couples the *UOB TMRW* self-service ecosystem with *Chat Assist*; and OCBC coordinates intent-driven chatbots with multi-channel live-agent handoffs **[7], [8], [9]**. In parallel, digital banks such as Revolut explicitly disclose the boundaries of automated AI assistance, publicize error fallbacks, and define explicit routing pathways to human advisors **[10].**

These industrial architectures illustrate that conversational banking cannot be framed simply as a dichotomy between rigid, rule-based trees and unconstrained large language models (LLMs). While predefined dialogue flows fail to accommodate interleaved or out-of-vocabulary user requests, unconstrained generative models lack deterministic authority to execute financial operations. To establish the functional scope of NovaBank, a structured review of 27 publicly available customer-service guides across DBS, UOB, OCBC, and Revolut was conducted. This benchmarking informed NovaBank's operational policies and state transitions without attempting to reverse-engineer closed proprietary banking engines.

3.2 Technical Review: Intent Recognition and Retrieval-Augmented Generation

Benchmarking fine-grained financial requests relies heavily on standardized datasets. Casanueva et al. (2020) introduced BANKING77, comprising 13,083 queries across 77 domain-specific intents **[11].** While BANKING77 provides an established evaluation baseline for intent discrimination, standard classification benchmarks operate under a closed-world assumption: high classification accuracy does not reflect whether a user query is semantically under-specified or whether the bank maintains sufficient policy documentation to resolve it. Consequently, NovaBank decouples statistical intent classification from downstream uncertainty governance, formalizing distinct handling branches for linguistic ambiguity (**CLARIFY**) and evidentiary insufficiency**(ABSTAIN) [11]**.

For domain knowledge grounding, Lewis et al. (2020) established Retrieval-Augmented Generation (RAG) to combine parametric language modeling with non-parametric corpus retrieval **[12]**. Murtaza et al. (2025) extended this paradigm to structured and unstructured banking workflows within financial contact centers **[13]**. However, off-the-shelf retrieval does not ensure that retrieved contexts satisfy all legal and procedural predicates governing an answer. In document preprocessing, fixed-window chunking risks decoupling conditional clauses (such as processing timelines and fee exemptions) from core policy statements. NovaBank therefore adopted topic-bounded semantic chunking, partitioning domain guidelines into 151 self-contained units with preserved conditionality.

At the retrieval stage, standard single-stage lexical search (BM25) is computationally efficient but sensitive to lexical mismatch. NovaBank enhances the lexical candidate pool by indexing structural document headings and 924 curated FAQ paraphrases, subsequently

deploying a cross-attention reranker (cross-encoder/ms-marco-MiniLM-L6-v2) for deep semantic relevance scoring, following established retrieve-and-rerank methodologies **[2]**, **[14]**. Nevertheless, system design must accommodate fundamental retrieval boundaries: rerankers cannot recover candidates missed by upstream stages, and high semantic relevance does not guarantee factual completeness.

3.3 Hybrid Reasoning Architectures and Auditable AI Governance

Integrating programmable guardrails alongside probabilistic language models is essential for safety-critical domains. Rebedea et al. (2023) presented NeMo Guardrails, demonstrating that conversational boundaries and programmable dialog rails can be defined independently of the generative model **[15]**. Rather than adopting third-party runtime dependencies, NovaBank implements a native, lightweight state machine and deterministic rule engine, directly tailoring stateful transitions to retail banking escalation protocols.

NovaBank's hybrid architecture establishes an inspectable boundary between probabilistic language processing and deterministic service control (risk escalation, complaint tracking, and recommendation gating). This design addresses practical governance without relying on complex, unproven neuro-symbolic learning paradigms. Its reliability naturally depends on the correctness of upstream classification signals and deterministic policy logic.

4 System Architecture and Model

4.1 System Overview

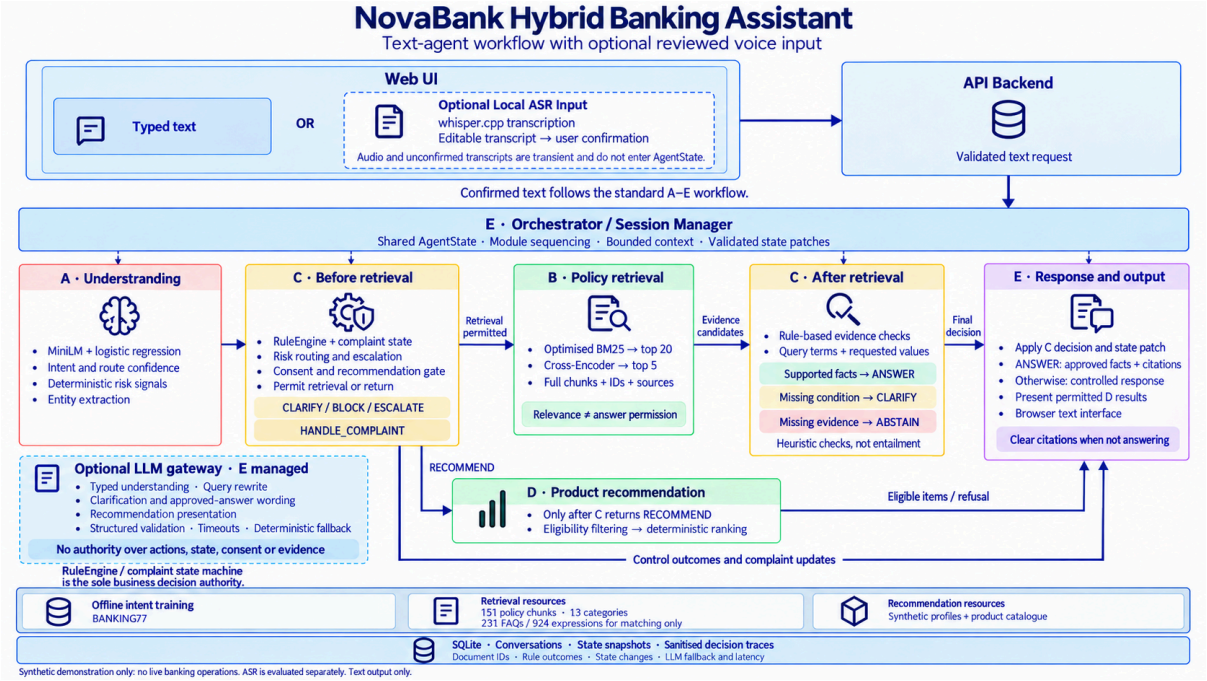


Figure 4.1 illustrates the functional modules, shared data resources, and orchestration boundaries of the NovaBank assistant.

NovaBank adopts a modular hybrid architecture that combines language processing, evidence retrieval, and deterministic business rules within a unified customer-service workflow. The design

separates the interpretation and expression of customer requests from decisions concerning permitted service actions. Machine learning supports intent classification, while deterministic risk detection and explicit business rules govern risk-sensitive routing, complaint handling, escalation, and recommendation permission.

The architecture separates offline data preparation from online request processing. Offline preparation provides the intent-classification data, curated NovaBank policy corpus, FAQ expression mappings and synthetic customer records required by the respective modules. The evaluated online workflow accepts text input and returns text responses. Its main functions are organised into four subsystems coordinated by a central orchestrator over shared conversation state:

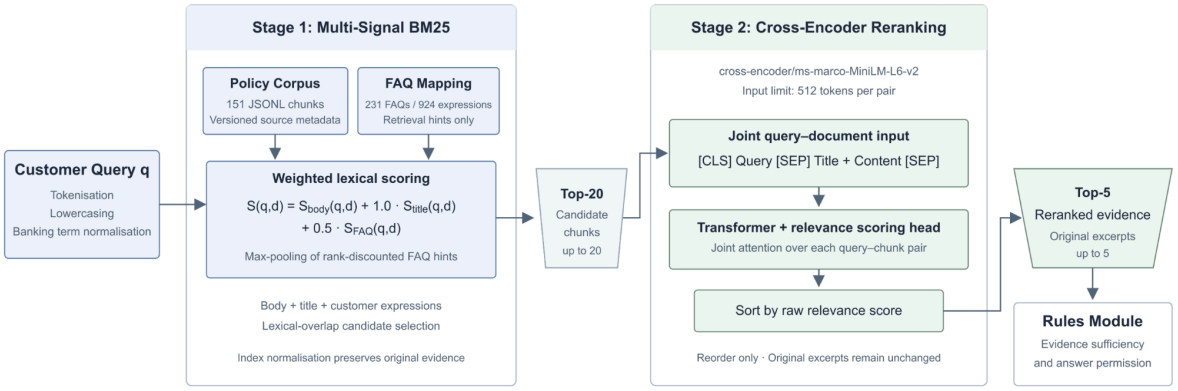
1. **Perception and Extraction** identifies the intent, risk signals and relevant entities in the current customer utterance.
2. **Reasoning and Control** applies deterministic business rules and complaint-state transitions to determine the permitted service action.
3. **Knowledge and Action** retrieves policy evidence through BM25 candidate selection and Cross-Encoder reranking, or filters and ranks eligible products after recommendation permission has been granted.
4. **Output Interface** presents the resulting response, citations and permitted recommendations to the user.

Structured decision traces record module outcomes, retrieved document identifiers, state changes and routing reasons to support post-hoc inspection. The workflow operates with synthetic customer profiles, fictional NovaBank policies and simulated handoffs, without accessing live banking accounts or executing financial transactions.

A separate local ASR component is available for producing candidate transcripts, but it is not part of the evaluated browser-based text workflow. TTS and integrated speech-to-speech interaction remain interface extensions. The architecture therefore distinguishes the implemented text-agent workflow from the additional integration and evaluation required for a complete voice interface.

4.2 Retrieval-Augmented Generation Model

The retrieval-augmented generation (RAG) component provides the banking assistant with identifiable policy evidence for answering customer enquiries. Its purpose is to retrieve relevant information from the fictional NovaBank knowledge base so that downstream responses can refer to documented service conditions rather than rely solely on a language model's internal knowledge. The retrieval module supplies evidence and source metadata; the rules module determines whether that evidence is sufficient for an answer, and the orchestrator assembles the final response and citations.



The implemented retrieval backend follows a two-stage approach. An optimised BM25 retriever first selects candidate knowledge chunks using lexical matching, title information, and alternative customer expressions. A Cross-Encoder then evaluates the relevance of each query–candidate pair and reranks the candidates. This combines an inspectable first-stage retrieval process with contextual relevance scoring while preserving the original evidence text.

4.3 Recommendation Decision Model

The product recommendation component selects and ranks demonstration products after the rules component has permitted the recommendation pathway. Module C determines whether the service context permits a recommendation, while Module D evaluates product eligibility and ordering against an explicitly associated synthetic customer profile. Module E coordinates invocation and state integration. This separation prevents product eligibility from being treated as sufficient authority to recommend during an unresolved complaint or without the required consent.

D accepts a session identifier, complaint status, handoff status and consent. The interface contains neither current risk information nor proof of C's decision, so E must invoke D only after receiving RECOMMEND from C. D does not resolve complaints, grant consent, assess conversational risk or execute account-opening actions. Its defensive checks can reject an invocation but cannot establish that the upstream permission conditions have been satisfied.

The recommendation design separates hard eligibility constraints from softer ordering preferences. Age, residency and available deposit determine whether a product can enter the candidate set; a stated goal then influences the order of eligible products. This makes the implemented decision model explicit and keeps ranking priorities from overriding eligibility. Figure D1 summarises the ownership boundary and selection stages.**[16]**

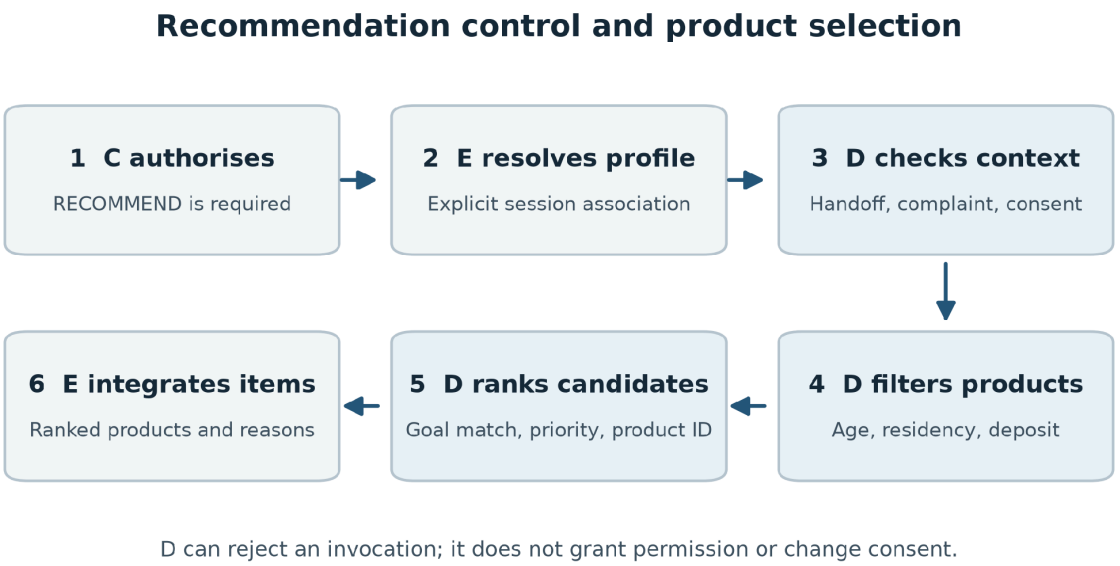


Figure D1 Recommendation permission and product selection. D performs defensive checks and product selection after C permits the pathway.

4.4 Intent, Risk and Entity Understanding

The intent, risk and entity understanding component forms the perception layer of NovaBank. It converts a customer utterance into a bounded semantic record that downstream rules can interpret consistently. Its input is the current session identifier and recognised user text; its output contains an intent label, a model-derived route confidence score, a risk level and extracted entities. The component does not select a service action, change complaint or handoff state, grant consent, retrieve policy evidence or compose the final response. These decisions remain with the reasoning, knowledge and output modules.

BANKING77 supplies 77 fine-grained banking intents and is used as the supervised classification benchmark. Because these labels are more specific than the operational vocabulary required by NovaBank, the implementation retains the 77-class classifier and applies an explicit route mapping above it. Nineteen fine-grained labels have direct route assignments. The remaining 58 labels default to **policy_enquiry**, which preserves an informational response pathway without implying permission for a consequential action. **UNKNOWN** is reserved for empty or non-informative input; it is not a BANKING77 class, and the trained classifier therefore always returns one of its 77 known labels. Low-confidence handling is performed by the surrounding rules rather than being represented as an additional learned class.**[17]**

Route	Operational purpose	Relationship to BANKING77
card_fee_enquiry	Questions about card fees, charges or billing rules	Explicitly mapped fee-enquiry labels

fee_complaint	Dissatisfaction concerning a fee or charge	Explicitly mapped complaint labels
unauthorized_transaction	Recognition, dispute or investigation of an unauthorised payment	Explicitly mapped transaction labels
fraud_suspicion	Suspected scam, account compromise or fraudulent activity	Explicit mappings supplemented by risk phrases
product_recommendation	Opt-in product guidance when service conditions permit	Not represented in BANKING77; permission is controlled by Module C and product selection by Module D
policy_enquiry	General account, card, payment and transfer policy questions	Explicit mappings plus the 58 labels without a specialist route
UNKNOWN	No usable intent can be inferred	Runtime fallback, not a BANKING77 label

Risk is assessed separately from intent confidence, using both phrase-level signals and an intent-derived risk prior. This allows explicit security signals to contribute to the risk assessment even when the predicted intent is uncertain. High-risk phrases include “without my permission”, “unauthorised”, “scam”, “fraud”, “stolen”, “hacked”, “suspicious” and “someone used”. Medium-risk phrases include “charged twice”, “overcharged”, “wrong amount”, “dispute”, “refund”, “declined”, “failed” and “pending”. A high-risk phrase cannot be downgraded by low intent confidence. The resulting level is a routing signal only; the rules module remains responsible for escalation and any final action.

Entity extraction identifies the information needed to interpret and contextualise a banking request without attempting to recover unnecessary personal data. The supported types are **payment_method**, **amount**, **currency**, **date**, **relative_time**, **account_type**, **product**, **channel** and **transaction_reference**. A gazetteer supplies controlled banking vocabulary, while regular expressions handle monetary values, dates and relative time expressions. Matching uses longest-phrase precedence so that a specific product name is not reduced to a generic account type. Sensitive or

uncertain references are handled conservatively and are not expanded into inferred personal attributes.**[18]**

The resulting contract is intentionally narrow: `{status, data: {intent, confidence, risk_level, entities}}`. It does not expose an action, reason code, state transition or response text. The separation allows the perception component to be tested as a classifier and entity extractor, while policy compliance, complaint handling, recommendation permission and final response generation remain auditable responsibilities of the other modules.

4.5 Rules and Complaint State Machine

Module C is the service-decision boundary between probabilistic interpretation and customer-facing action. It receives the shared AgentState, including Module A's intent, confidence, and risk signals; the current complaint, handoff, and consent states; any bounded conversational reference supplied by Module E; and, after retrieval, Module B's policy candidates. It returns a structured Decision containing an action, a reason_code, and a restricted next_state patch. The available actions are **ANSWER, HANDLE_COMPLAINT, ESCALATE, RECOMMEND, CLARIFY, BLOCK**, and **ABSTAIN**. The patch may change complaint or handoff status; it cannot grant consent, alter Module A's interpretation, select products, or write the final response. Module E applies valid patches and coordinates subsequent modules.

Table C1 Decision order and service-action boundaries

Order	Decision boundary	Result
1	Unauthenticated account read, transfer, or payment execution request	BLOCK; these operations are unavailable in the prototype
2	Explicit urgent report, urgent intent or high-risk signal, provided that no earlier operation-blocking rule has matched	ESCALATE, with a narrow exception for a classifier-only high-risk label on a contextual fee dispute
3	Recommendation with unresolved service, active handoff, missing consent, or non-low risk	BLOCK before Module D is called
4	Active handoff or explicit or contextually resolved complaint	ESCALATE or HANDLE_COMPLAINT, as appropriate
5	Missing task context, low confidence, unknown intent, or unknown risk	CLARIFY
6	Recommendation that passes all service conditions	RECOMMEND; D still determines product eligibility and order
7	Ordinary policy enquiry	Permit retrieval, then ANSWER, CLARIFY, or ABSTAIN after evidence inspection

The ordering matters because Module A supplies signals rather than authority. A clear report that a card was used without permission must not become an ordinary policy answer merely because its predicted intent is uncertain or generic. Conversely, a request to read a personal balance cannot be satisfied by retrieving general policy text. For recommendations, an **OPEN** or **UNDER_REVIEW** complaint blocks the request. Permission requires complaint status NONE or RESOLVED, no active

handoff, GRANTED consent, and LOW risk. Module C's **RECOMMEND** action authorises Module E to call Module D; it does not certify that any product is eligible or suitable.

Module C distinguishes two reasons for withholding an answer. **CLARIFY** applies when the customer's task or a required condition is missing, including a replacement-card question whose applicable fee depends on the reason for replacement. **ABSTAIN** applies when the request is sufficiently understood but admissible evidence is absent or does not support the requested fact. A documented policy restriction can itself support a bounded negative answer; lack of documentation cannot.

Complaint handling uses separate complaint and handoff states. An explicit complaint opens a simulated case or preserves an existing **OPEN** or **UNDER_REVIEW** case. An urgent security report opens a case when necessary and marks the handoff **ESCALATED**; it does not resolve an existing case. The case-transition validator permits **NONE** or **RESOLVED** to **OPEN**, **OPEN** to **UNDER_REVIEW**, and **UNDER_REVIEW** to **RESOLVED** only with the required actor, revision, resolution evidence reference, and handoff clearance. A customer statement that a complaint is resolved is not, by itself, a valid human resolution event. These transition rules define the control boundary; the present prototype does not provide a complete authenticated human case-management workflow.

4.6 LLM Assisted Agent Safety Boundary

The LLM supports semantic understanding and response delivery, but it cannot alter the RuleEngine decision, complaint or handoff state, consent, evidence approval, or recommendation eligibility. The permitted LLM stages and their safety boundaries are summarised below.

LLM stage	Trigger	Function	Safety boundary
Typed understanding and fuse	Feature flag enabled	Provides a structured intent, risk and entity draft.	Cannot choose an action or weaken an existing safety signal.
Query rewrite	C initially returns ANSWER	Creates an additional retrieval query.	Preserves the original query and cannot permit retrieval.
Clarification wording	C returns CLARIFY	Produces a focused follow-up question.	Cannot decide whether clarification is needed or change state.
Grounded answer wording	C approves evidence-backed ANSWER	Generates final answer wording from approved evidence.	Cannot add unsupported facts or citations.
Recommendation presentation	C permits and D returns items	Produces a non-factual introduction.	Cannot change product names, ranks, reasons or eligibility.

4.7 Optional Voice Input Architecture and Safety Boundary

The evaluated NovaBank workflow is text-based; however, the prototype includes an optional local automatic speech recognition (ASR) boundary to support voice input. The voice component is positioned before the ordinary text-agent workflow rather than being embedded within the intent, retrieval or rules modules. Its sole function is to produce a candidate text transcript from a user recording. The transcript is then reviewed and may be edited by the user before it is submitted as a normal text request.

This separation ensures that speech recognition does not receive authority to determine intent, risk, complaint status, recommendation eligibility or service actions. Once the user confirms the transcript, the request enters the same Module A to E workflow as typed text and is subject to the same retrieval, rules and safety controls. An unconfirmed transcript therefore has no effect on shared conversation state.

The voice boundary follows a data-minimisation approach. Audio is processed locally and is not persisted in the conversation database, AgentState or decision trace. Similarly, unconfirmed transcripts are not stored as chat messages. The prototype currently provides speech-to-text input only; text-to-speech and a fully evaluated speech-to-speech banking workflow remain outside the evaluated system scope.

5 System Development and Implementation

5.1 BANKING77 Dataset and Preparation

BANKING77 provides the supervised data for NovaBank's intent-classification component. Released by PolyAI and introduced by Casanueva et al. **[11]**, it contains **13,083 English online-banking customer-service queries** annotated with **77 fine-grained intents**. Each instance pairs a single-turn utterance with one intent label. Topics include lost or stolen cards, unrecognised payments, pending transfers, refunds and PIN changes. Closely related categories, such as `pending_transfer` and `transfer_not_received_by_recipient`, make the dataset suitable for examining distinctions between semantically similar service requests.

The project preserves the official split of **10,003 training examples** and **3,080 test examples**. A **12% stratified development holdout**, generated from the training partition with random seed **42**, yields **8,802 fitting examples** and **1,201 development examples**. Stratification preserves the relative class distribution during model selection. After selecting the model and regularisation configuration using development Macro-F1, the classifier is refitted on the complete official training partition. The test partition remains separate from model selection and hyperparameter tuning.

Partition	Examples	Purpose
Fitting subset	8,802	Fit candidate classifiers
Development subset	1,201	Select models and hyperparameters

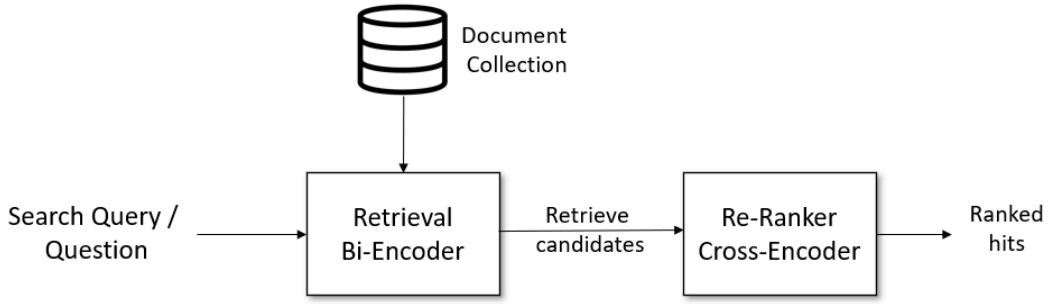
Partition	Examples	Purpose
Complete official training partition	10,003	Refit the selected classifier
Official test partition	3,080	Final benchmark evaluation

The fitting and development subsets together constitute the official training partition; they are not additional data. For the selected frozen-encoder configuration, utterances retain their original case and punctuation, without stopword removal. BANKING77 trains the classification layer over pretrained sentence embeddings; it is not used to fine-tune the encoder end to end. Predictions retain the **77-class** label space for evaluation and are mapped to NovaBank's operational routes for downstream processing. The mapping assigns **19 labels** to specialist routes and the remaining **58 labels** to `policy_enquiry`.

BANKING77 supplies intent annotations rather than policy facts, multi-turn histories, slot-filling annotations or structured transaction parameters. It also lacks a dedicated product-recommendation intent and an out-of-scope class. Its role is therefore limited to intent learning and routing evaluation; business permissions and workflow behaviour require separate rules and scenario tests.

5.2 Retrieval-Augmented Generation

5.2.1 Knowledge Base Construction



General retrieve-and-rerank pipeline. NovaBank uses BM25 for candidate retrieval and a Cross-Encoder for reranking. Source: Sentence Transformers documentation.

The NovaBank knowledge base was developed using public banking guidance as a reference for authoring fictional service policies. A reference register records **27 official web pages** from DBS, UOB, OCBC and Revolut, covering card management, transaction disputes, account access, identity verification and refunds. These materials informed distinctions such as temporary card freezing versus permanent cancellation, and pending authorisations versus settled disputes. NovaBank-specific fees, limits, processing periods and liability conditions were defined as synthetic project parameters rather than adopted as verified terms of the referenced institutions.**[19]**

The NovaBank corpus contains **151 policy chunks** organised into **13 banking categories** and mapped to BANKING77 intent labels. Policies were grouped by topic to preserve relevant conditions, exceptions and processing dependencies within each chunk. Each JSONL record stores the policy text, a unique identifier, version information, category and intent mappings. Only NovaBank policy chunks

are returned as answer evidence. External reference pages are excluded from the evidence corpus, while FAQ expressions are indexed separately to support candidate retrieval.

5.2.2 FAQ Expression Mapping

Customer queries frequently rely on conversational vocabulary that diverges from codified policy terminology (e.g., asking how to "get another card" rather than requesting "card replacement"). To bridge this lexical gap without mutating source policies, the system implements an auxiliary mapping between colloquial question expressions and candidate policy chunks.

The mapping comprises 231 distinct FAQ entries expanded into 924 expressions covering primary questions and alternative syntactic formulations. Each FAQ candidate mapping requires matching policy identifiers, versions, and categories, combined with an explicit connection to a chunk's primary or secondary intent. Crucially, this layer acts strictly as a discovery signal for first-stage lexical matching: FAQ expressions never serve as factual authority, and alternative phrasing creates no redundant copies of the underlying ground-truth chunks.**[20]**

5.2.3 BM25 Candidate Retrieval

The initial retrieval stage scores candidate chunks using an optimized BM25 ranker. Queries and corpus texts undergo identical preprocessing: lowercasing, tokenization, conservative stopword filtering, and banking-specific inflection normalization (e.g., normalizing "fees" and "cards", and mapping "replacing" to "replace"). Numerical entities, decimal amounts, and negation markers are explicitly preserved to safeguard financial limits and policy conditions.

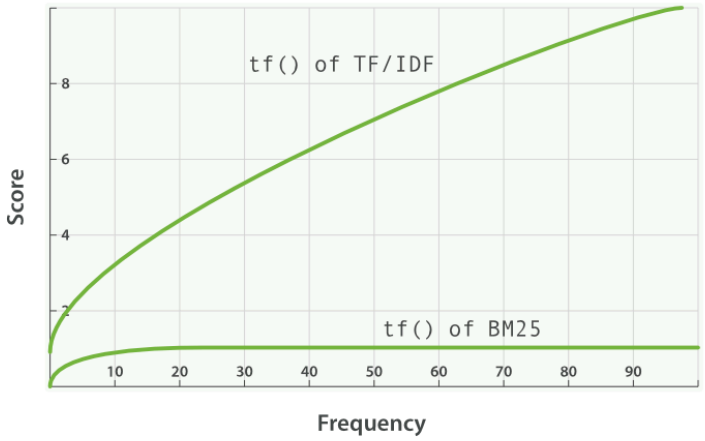


Illustration of term-frequency saturation in BM25 compared with TF-IDF. Source: Elastic.

The retriever integrates signals across chunk bodies, titles, and associated FAQ expressions:

$$S(q,d) = S_{body}(q,d) + w_{title} \cdot S_{title}(q,d) + w_{faq} \cdot S_{faq}(q,d)$$

where q denotes the query, d is the candidate chunk, and the default weights are configured to $w_{title} = 1.0$ and $w_{faq} = 0.5$. To prevent multi-mapped chunks from accumulating disproportionate

bias, FAQ contributions are discounted by the chunk's positional rank within the candidate mapping, and max-pooling is applied across multiple matching expressions.

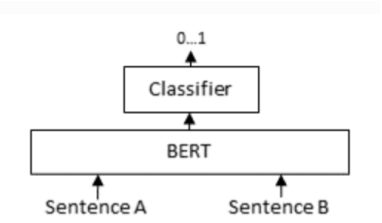
Only chunks sharing indexed vocabulary with the query remain eligible. Candidates are ordered by aggregate score, with persistent document identifiers providing deterministic tie-breaking. By default, the top 20 candidate chunks are passed to the second stage.

5.2.4 Cross-Encoder Reranking

The second stage deploys cross-encoder/ms-marco-MiniLM-L6-v2 to perform contextual reranking over the top 20 candidates. For each candidate, the model jointly encodes the query and the chunk's title and body using full cross-attention, capturing deep semantic relevance beyond lexical co-occurrence.

The model revision is strictly pinned to ensure evaluation reproducibility. Query–candidate pairs are processed in mini-batches, and the top 5 highest-scoring chunks are emitted. The reranker updates relevance scores and ordering without mutating underlying content, source metadata, or document identifiers.

Because Cross-Encoder logits represent raw relevance scores rather than calibrated probabilities of factual sufficiency, no heuristic rejection threshold is enforced at this stage.



Instead, downstream sufficiency checks and fallback actions remain strictly governed by the reasoning module (Module C). Truncation is handled gracefully against the model's 512-token context window, while emitted evidence retains the complete, uncompressed source chunk.

5.3 Product Recommendation

5.3.1 Synthetic Data and Validation

The recommendation fixture contains four fictional customer profiles and three demonstration products. Customer records specify age, residency, available deposit and a stated goal. Product records specify minimum age, supported residencies, minimum deposit, supported goals and a demonstration priority. The profiles include an eligible adult, an underage customer, a customer outside the supported residency condition and an adult with a small available deposit. They are controlled test cases rather than observations of real customer behaviour.

Strict validation rejects unknown fields, blank identifiers, duplicate product identifiers and malformed values. Age and monetary amounts must be nonnegative integers; booleans, floating-point values and numeric strings are not implicitly converted. Amounts are stored in integer SGD cents so that threshold comparisons do not depend on floating-point rounding. Product

condition lists must be nonempty, and the fixture must explicitly identify itself as synthetic. Missing or invalid profile data are never replaced with a default customer.**[21]**

Table D1 Synthetic product conditions. All three products support residency SG; values are demonstration parameters, not approved bank-product terms.

Product	Min age	Min deposit SGD	Supported goals	Priority
Demo Flexible Savings	18	0.00	savings; emergency	20
Demo Premium Savings	21	1,000.00	savings	10
Demo Emergency Reserve	18	200.00	emergency	5

5.3.2 Eligibility and Deterministic Ranking

Eligibility is evaluated before ranking. A product enters the candidate set only when the customer's age meets its minimum age, the customer's residency belongs to its supported residency set and the available deposit meets its minimum deposit. Consequently, a favourable ranking priority cannot compensate for a failed eligibility condition. The customer's stated goal affects ordering rather than eligibility.

Eligible products are sorted lexicographically by three criteria: goal mismatch, ascending demonstration priority and product identifier. Goal-matching products receive a mismatch value of zero and precede alternatives with a value of one. Within either group, smaller demo_priority values rank first, and product identifiers provide a deterministic tie-breaker. The priority is an explicit demonstration parameter, not a learned preference score, expected financial return or measure of product suitability.**[22]**

For the adult SG-resident profile with SGD 1,500 available deposit and a savings goal, the order is Demo Premium Savings, Demo Flexible Savings and Demo Emergency Reserve. Emergency Reserve has the smallest numeric priority but follows the two savings-matching products because goal matching takes precedence. Changing the same profile's goal to emergency produces Emergency Reserve, Flexible Savings and Premium Savings.

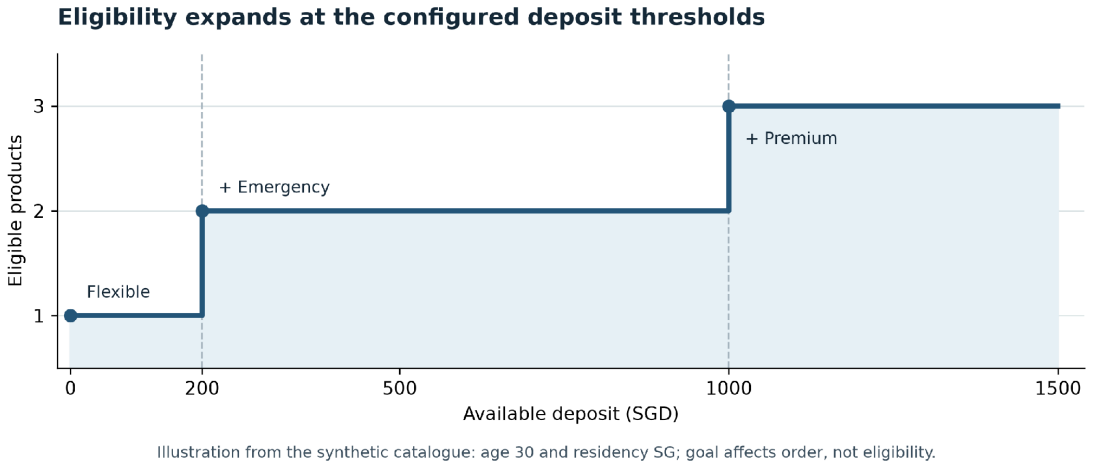


Figure D2 Deposit thresholds in the demonstration catalogue. Eligibility is illustrated for age 30 and residency SG; this is derived from configured rules rather than an empirical performance result.

Each item contains a product identifier, name, consecutive rank and explanation. The explanation identifies satisfaction of the implemented age, residency and deposit conditions and states whether the product matches the synthetic goal. An eligible alternative is explicitly described as not matching the goal where applicable. These explanations make the implemented selection rule inspectable without implying an optimal financial choice.

5.3.3 Defensive Outcomes and Integration

Expected business refusals are distinguished from technical failures. With no active handoff, an OPEN or UNDER_REVIEW complaint produces UNRESOLVED_COMPLAINT, and unknown or denied consent produces CONSENT_REQUIRED. Complaint rejection precedes consent rejection. A valid catalogue with no eligible products produces NO_ELIGIBLE_PRODUCT. These results use status=ok, allowed=false and an empty item list; a valid empty candidate set is not treated as a malfunction.

An active handoff, malformed request, invalid catalogue or unknown session/profile association produces a technical error without recommendations. These defensive checks supplement C's service-level gate. D's acceptance of complaint status NONE or RESOLVED does not replace C's recommendation permission checks on complaint state, handoff, consent, risk and confidence.

The implementation separates a synchronous selection function from an asynchronous adapter compatible with E's recommendation interface. The adapter validates and snapshots injected configuration and returns independent result objects. Recommendation calls perform no file or network I/O; a separate loader reads an explicit trusted fixture path during initialisation. Expected technical errors are adapted to E's typed result envelope, while invalid startup configuration and unexpected failures remain subject to the application's safe error-handling path. D returns product-level outcomes, and E integrates the items without changing C's decision.

5.4 Intent, Risk and Entity Understanding

5.4.1 Data Preparation and Operational Label Design

Following the dataset preparation protocol in Section 5.1, fine-grained predictions are converted into the operational routes defined in Section 4.4. The mapping is stored as a versioned artifact and checked for complete label coverage and valid route values. Mapping a prediction to a route supplies an interpretation for downstream processing; it does not authorise a service action or modify complaint, consent or handoff state.

5.4.2 Classifier Training with a Frozen Sentence Encoder

The selected model uses the **sentence-transformers/all-MiniLM-L6-v2** encoder at revision `1110a243fdf4706b3f48f1d95db1a4f5529b4d41` under the Apache-2.0 licence. Each utterance is encoded as a 384-dimensional, L2-normalised sentence embedding with a maximum sequence length of 256 and a batch size of 64. A multinomial logistic-regression classifier is fitted on the embeddings using `C=10.0`, `class_weight="balanced"`, `max_iter=2000`, `random_state=42` and the L-BFGS solver.

Candidate models were evaluated on the development split using Macro-F1 for selection, while the test split was kept strictly isolated from model selection and hyperparameter tuning. Following selection, the logistic-regression classifier was refitted on embeddings from the complete training

split, keeping the pretrained sentence encoder frozen. During inference, the resulting 77-class classifier computes class probabilities via $softmax(eW^T + b)$. The highest-probability fine-grained label is selected first and mapped to its operational route. The confidence score is the sum of the probabilities of all fine-grained labels mapped to that selected route.**[2], [14]**

5.4.3 Runtime Inference, Risk and Entity Processing

The published artifact contains the logistic-regression coefficients and intercepts, the 77-label order, the complete route mapping, the preprocessing configuration and a manifest with the pinned encoder revision and SHA-256 checksums. Encoder weights are not committed to the repository because the snapshot is approximately 90 MB. Instead, the immutable Hugging Face revision and per-file checksums freeze the external model snapshot, while the four local artifact files are verified at load time. Runtime inference uses NumPy softmax rather than requiring scikit-learn. Verification against the scikit-learn implementation produced no prediction mismatches on the test split and a maximum probability difference of approximately 2.915×10^{-6}.

Confidence is reported as the probability mass of the selected operational route rather than only the probability of the top fine-grained class. This is more appropriate for the downstream workflow because several BANKING77 labels may map to the same route and because the 77-way maximum can remain modest even when the operational destination is clear. The trained classifier always returns one of its learned labels; it does not emit an abstention class. Low-confidence handling, clarification and escalation are therefore retained by the rules module, where thresholds can be audited alongside the rest of the service policy.**[24]**

Risk detection and entity extraction remain deterministic. Risk combines the maximum of phrase-level evidence and the intent prior, preserving high-risk phrases even when the predicted intent is uncertain. Entity extraction uses a controlled gazetteer and regular expressions rather than an additional trained model. Gazetteer phrases are matched longest-first so that a specific expression such as “credit card” is not also duplicated as a generic “card”. Gazetteer and pattern matches carry explicit confidence values of 0.90 and 0.85 respectively. Amount, date and account-reference patterns are intentionally conservative because those values may require confirmation before any downstream action.

The asynchronous adapter accepts `session_id` and `user_text` and returns the typed envelope `{status, data: {intent, confidence, risk_level, entities}}`. The session identifier is accepted for interface compatibility but is not used to read or write conversation state. The module performs no policy retrieval, complaint lookup, consent check or recommendation call, preserving a clear separation between understanding and business authority.

5.5 Rules and Complaint State Machine Implementation

The implementation separates ordinary routing, complaint arbitration, evidence arbitration, and case-transition validation. RuleEngine produces the structured decision and enforces the decision order in Table C1. ComplaintArbitrator checks explicit urgent language independently of Module A’s label, handles complaint opening and escalation, and preserves unresolved case state. ComplaintStateMachine validates case events using an expected revision and event identifier; review and resolution require a human-agent actor, and resolution requires an evidence reference.

Authentication of that actor, durable storage of the full event history, and enforcement of globally unique event identifiers remain responsibilities of an integrating case-management service.

A policy question follows two Module C checks. Before retrieval, C decides whether an answer route is permissible. Module E then requests up to five ranked candidates from B. C removes malformed or duplicate candidates, retains B's order for ordinary enquiries, and performs a final evidence check. At least one admissible document must cover the salient terms of the question; a requested price or time also requires a corresponding value in the relevant passage. This is a deliberately bounded heuristic, not proof of full semantic entailment. Module E clears evidence and citations when C ultimately declines to answer, and invokes the grounded response generator only after an ANSWER decision.

The distinction is visible in replacement-card cases. A generic replacement-fee question can be answered if the retrieved policy states a genuinely generic fee. When the available policy instead gives different terms for different replacement reasons, C returns CLARIFY / REPLACEMENT_REASON_REQUIRED. A delivery period or express-shipping charge cannot establish the replacement fee, so a question supported only by those values receives ABSTAIN / INSUFFICIENT_EVIDENCE. An explicit lost-card report takes the security route, including when phrased as a fee question: the current implementation returns ESCALATE / CARD_SECURITY_RISK. A reviewed document that explicitly states the bank does not support cryptocurrency can, by contrast, support a limited negative answer about crypto-wallet recovery; the system must not invent recovery steps.

For recommendation requests, C checks the service state before D is called. An open complaint produces BLOCK / UNRESOLVED_COMPLAINT even when consent is present or Module A has assigned a different intent. When C permits the pathway, E passes the session association and state to D, which applies its own defensive checks, product eligibility conditions, and deterministic ranking. Optional language-model assistance may help interpret a bounded follow-up or phrase the eventual response, but it does not grant recommendation permission or replace C's action. The implemented runtime persists aggregate conversation state and sanitised decision traces; the case-transition validator should not be described as a completed human-review or evidence-backed resolution service.

5.6 LLM Gateway and Guarded Delivery Implementation

The integrated delivery pipeline uses a server-side OpenAI-compatible LLM gateway. Provider credentials remain on the server and are never exposed to the browser. The gateway provides typed request and response handling, structured-output validation, bounded retries, timeout controls, concurrency limits and request-level observability.

LLM assistance is applied at five controlled stages: typed understanding and conservative fusion, query rewriting, clarification wording, evidence-grounded answer wording and recommendation presentation. Query rewriting is invoked only after Module C has initially permitted the ANSWER pathway and before retrieval. The rewritten query is appended to the original customer wording so that financial amounts, time conditions, negation and other retrieval-critical details are preserved. The clarification generator is invoked only after Module C has already returned CLARIFY; it converts the approved clarification reason into a focused customer-facing question and cannot decide whether clarification is required.

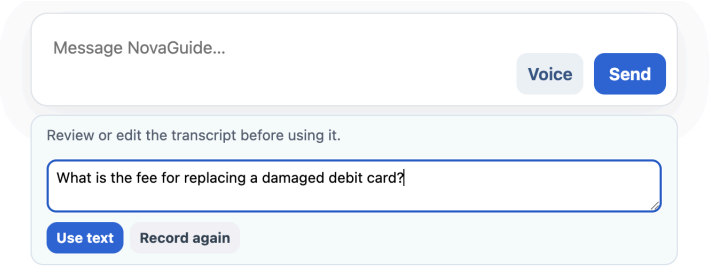
The grounded answer generator receives only evidence approved by Module C, while the recommendation presentation generator cannot alter Module D's product names, ranking, reasons or eligibility decisions. The LLM is prohibited from selecting a RuleEngine action, changing complaint, handoff or consent state, granting recommendation permission, approving evidence or executing financial operations. Before provider transmission, requests containing passwords, PINs, one-time passwords or CVV/CVC values are blocked, and recognised personal or financial identifiers are replaced with request-local placeholders. A timeout, invalid structured response, provider failure or budget breach triggers a controlled deterministic fallback. Sanitised traces record the LLM stage, validation result, fallback status, latency and token usage without retaining sensitive customer content.

5.7 Local ASR Implementation

The optional ASR implementation uses whisper.cpp with a fixed local Whisper large-v3-turbo-q5_0 model. The browser records a short microphone input and prepares a 16 kHz, mono PCM WAV file for submission to the local application. The server validates the audio format, duration and size before invoking the ASR backend. The initial implementation accepts a maximum recording duration of 60 seconds and a maximum upload size of 10 MiB.

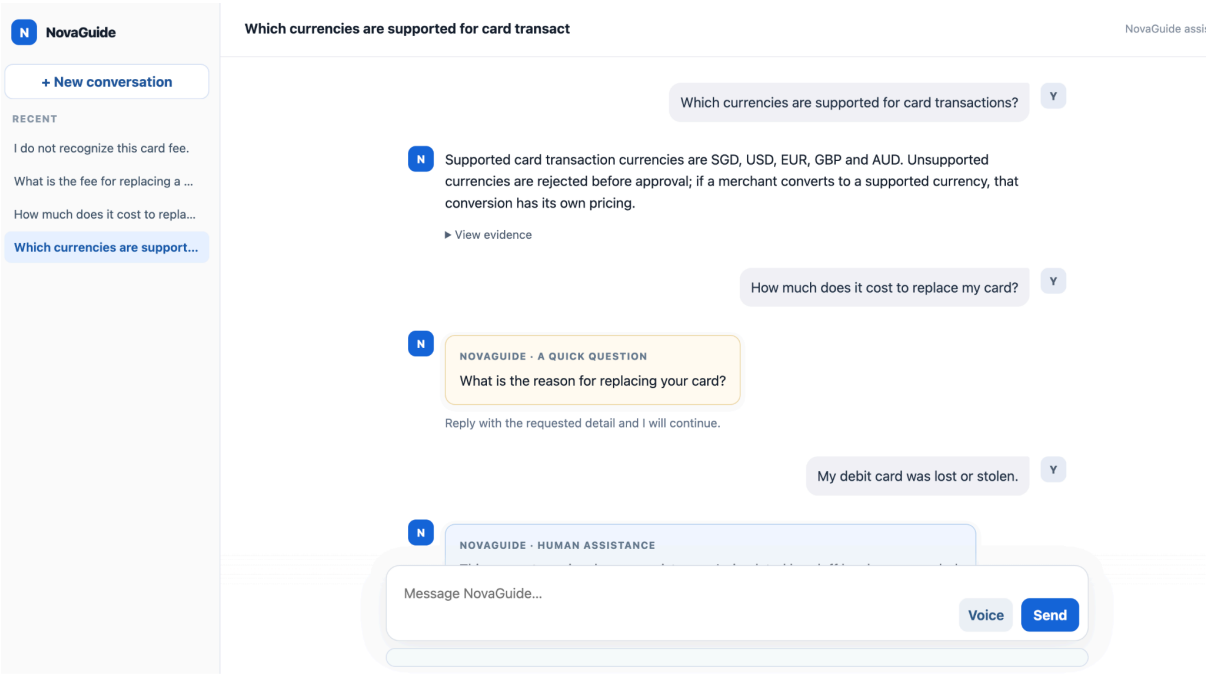
The transcription result is returned to the browser as editable text. The user must explicitly select the reviewed transcript before it is copied into the normal chat input field and sent to the banking assistant. If transcription is unavailable, invalid or empty, the system returns a controlled error and does not create a conversation turn. This confirmation step is important because ASR output may contain recognition errors, especially for financial terminology, names, amounts or mixed-language speech.

The ASR component runs through the local executable and pinned model described above. It is not used in the main text-agent evaluation reported in Chapter 6. Consequently, the reported intent-classification, retrieval, recommendation and rules results should not be interpreted as end-to-end voice-service performance. Further work should evaluate transcription quality, noisy recordings, mixed-language speech and the effect of transcript errors on downstream service safety.



5.8 Web Application and Local Persistence

The prototype is delivered through a lightweight local web application. The frontend is implemented using HTML, CSS and vanilla JavaScript, providing account registration and sign-in, conversation creation, chat-based text input, conversation-history display, evidence citation display and recommendation presentation. This dependency-light interface communicates with the application through same-origin JSON endpoints and is intended only for local synthetic-data demonstrations.



The backend is implemented in Python and served through the standard WSGI interface. It coordinates authentication, conversation handling, module orchestration, response formatting and controlled error handling. Pydantic-based schemas validate structured data exchanged between components and ensure that only permitted public response fields are exposed to the browser.

SQLite is used as the local persistence layer through Python's sqlite3 library. It stores synthetic user accounts, salted password hashes, session records, conversations, public chat messages, conversation-state snapshots and sanitised decision traces. The database contains no real banking records and is not intended as a production deployment datastore. This design keeps the demonstration self-contained while preserving clear boundaries between the user interface, application logic and stored synthetic data.

6 Experiment Results and Evaluation

This chapter evaluates NovaBank at component and integration levels. Sections 6.1–6.3 assess policy retrieval, product recommendation and intent classification, respectively, while Section 6.4 examines rule-based decisions, recommendation permission, evidence grounding and complaint-state continuity. Each evaluation uses task-specific datasets, scoring units and criteria. Table 6.1 summarises these protocols to clarify the scope of the reported results. Component performance and controlled workflow outcomes are interpreted separately rather than combined into a single measure of service quality.

Evaluation	Data and scoring unit	Configuration and assessment
Policy retrieval	71 evidence-bearing development questions; query-level metrics	BM25 retrieves up to 20 candidates; Cross-Encoder returns the top 5; compare results with approved required chunks
Product recommendation	Synthetic profiles and products; 44 test methods	Validate eligibility, deterministic ranking, defensive outcomes and adapter behaviour

Evaluation	Data and scoring unit	Configuration and assessment
Intent classification Official	BANKING77 test set; 3,080 utterances	Frozen MiniLM encoder with logistic regression; report fine-grained and mapped-route performance
Rules regression	31 decision-matrix cases and a separate 78-test suite**[25]**	Check actions, reason codes and complaint-state behaviour
Integrated workflows Reviewed	scenario fixtures; attempts, turns and sequences	Assess recommendation permission, grounding and complaint continuity under recorded freeze configurations

Table 6.1 Overview of evaluation protocols

6.1 RAG Retrieval Evaluation

The retrieval pipeline was evaluated using optimised BM25 candidate retrieval followed by Cross-Encoder reranking. Under the default configuration, up to 20 candidates were retrieved and reranked to return the five highest-scoring evidence chunks. The evaluation examined whether these results contained the policy passages required to answer each question, rather than assessing the fluency or factual accuracy of generated responses.**[26]**

The retrieval analysis covered 71 human-approved, evidence-bearing development questions. These comprised 62 questions concerning supported services and nine questions for which the policies explicitly stated that a requested service was unsupported. Both groups required policy evidence: a documented restriction can support a correct negative answer. Six questions required multiple knowledge chunks, allowing the evaluation to distinguish partial retrieval from complete evidence coverage.

Metric	Result	Interpretation
Hit@1	78.87%	First-ranked chunk contains at least one required evidence item
Recall@3	90.14%	Average proportion of required evidence chunks retrieved within three results
Recall@5	95.07%	Average proportion of required evidence chunks retrieved within five results
Hit@5	95.77%	At least one required chunk appears within five results
MRR@5	0.8531	Reciprocal rank of the first relevant chunk, averaged across questions

Metric	Result	Interpretation
Complete evidence retrieval for multi-chunk questions	5/6	All required chunks were retrieved within five results

Recall, Hit, and MRR were calculated over the 71 evidence-bearing questions. Complete multi-chunk retrieval was calculated over the six questions requiring multiple chunks.

Recall@k measures the proportion of required evidence chunks retrieved within the first *k* results, averaged across questions. Hit@k records whether at least one required chunk appears within that range. These measures differ when an answer requires multiple passages: retrieving one relevant chunk counts as a hit but may provide only partial support. MRR@5 averages the reciprocal rank of the first relevant result, assigning zero when no relevant result appears within the first five positions.

The pipeline achieved a Recall@5 of 95.07% and a Hit@5 of 95.77%, indicating that the returned set usually contained the expected policy evidence. Hit@1 was lower at 78.87%, showing that the highest-ranked passage alone did not consistently identify the required material. Recall increased from 90.14% at three results to 95.07% at five, supporting the inclusion of several passages for downstream inspection rather than exclusive reliance on the first result.

Performance varied between question types. Recall@5 reached 97.58% for supported-service questions but 77.78% for questions requiring evidence of an unsupported service. Although the latter group contained only nine questions, the difference identifies restriction-sensitive retrieval as an area requiring further attention. A passage may discuss the correct topic without establishing whether the requested service is permitted. Complete evidence was retrieved for five of the six multi-chunk questions, demonstrating that high aggregate recall does not guarantee coverage of every required condition.

Overall, these results support the pipeline's ability to retrieve policy evidence within the evaluated development setting. They do not independently establish evidence sufficiency, generated-answer correctness, or end-to-end safety. As the frozen test set had not yet been evaluated, the findings do not establish held-out generalisation or superiority over alternative retrieval configurations. Further validation should examine policy restrictions and questions requiring combined evidence across multiple chunks.

6.2 Recommendation Module Validation

Validation checks eligibility, ranking and rejection against independently specified expected outcomes. In local main-snapshot testing on 20 October 2026, all 41 D-specific unit-test methods and three D/shared-state integration methods passed without skips (Table D2). These 44 methods exclude four additional shared State/schema methods included in the earlier 48-method branch validation.

Table D2 Recommendation validation methods

Validation area	Methods	Recorded result
Synthetic data validation	10	Passed

Validation area	Methods	Recorded result
Eligibility and ranking	8	Passed
Defensive behaviour and integrity	15	Passed
Async adapter and result isolation	8	Passed
Shared-state integration	3	Passed
Total D-related methods	44	Passed

Boundary tests examine values immediately below, at and above product thresholds. For Demo Premium Savings, these include ages 20, 21 and 22, and deposits of 99,999, 100,000 and 100,001 cents. Ranking tests cover all six permutations of the three-product catalogue, repeated calls, goal-match precedence and identifier-based tie-breaking. Adapter tests examine configuration snapshots, concurrent calls and independence of returned values.

Table D3 Representative recommendation outcomes

Case	Synthetic input or condition	Observed D outcome
Permitted savings case	Adult SG profile; SGD 1,500; RESOLVED; GRANTED; no active handoff	Premium, Flexible, Emergency; ranks 1 to 3
Partial eligibility	Adult SG profile; SGD 10; otherwise valid context	Flexible only
Underage or unsupported residency	Age 17, or residency OTHER; otherwise valid context	NO_ELIGIBLE_PRODUCT; empty items
Unresolved complaint	OPEN or UNDER_REVIEW; no active handoff	UNRESOLVED_COMPLAINT; empty items
Consent not granted	RESOLVED; UNKNOWN or DENIED consent; no active handoff	CONSENT_REQUIRED; empty items
Unknown profile or invalid catalogue	Unrecognised session, duplicate product ID or malformed condition	Technical error; no default-customer fallback

All six catalogue input-order permutations produced the same result in the recorded regression test. Shared-state integration verified that returned items could be serialised and merged while preserving C's decision and the complaint and consent state. The recorded outcomes support functional correctness on the tested fixtures and boundaries; they do not measure customer satisfaction, financial suitability or population-level safety.

Negative calls verify D safeguards. Integrated testing must separately confirm that C blocks prohibited invocations. These module tests do not establish completion of the API/UI workflow. Controlled cross-module results are reported separately.

6.3 BANKING77 Intent Classification Evaluation

Using the dataset preparation protocol described in Section **5.1** and the classifier-training procedure described in Section **5.4.2**, the selected model was evaluated on the **3,080-example** official BANKING77 test partition. Macro-F1 is reported as the primary metric, with accuracy and mapped operational-route performance providing complementary views of classification performance. **[27]**

Table A2 Selected intent model results

Metric	Result	Interpretation
Test Macro-F1	0.9307	Unweighted performance across all 77 intents
Test accuracy	0.9312	Proportion of test utterances assigned the correct fine-grained label
Development Macro-F1	0.9213	Selection result before refitting on the full training split
Operational-route Macro-F1	0.9604	Performance after mapping fine-grained labels to routes
Operational-route accuracy	0.9838	Proportion of test utterances assigned the correct route

Route-level performance is more directly relevant to the agent than 77-way accuracy because multiple fine-grained labels can lead to the same downstream service pathway. Five routes are represented in the BANKING77 test labels and were therefore scored. **product_recommendation** is absent from the dataset and is not included in the route calculation.

Table A3 Route-level test results

Route	F1	Support
card_fee_enquiry	0.9645	240
fee_complaint	0.9694	280
fraud_suspicion	0.9426	120
policy_enquiry	0.9922	2320
unauthorized_transaction	0.9333	120

The **policy_enquiry** route aggregates **58 labels** and accounts for **2,320 of the 3,080 test examples**; the fraud and unauthorised-transaction routes each contain **120 examples**. Errors between labels mapped to the same route do not reduce route-level accuracy. Consequently, the mapped results measure coarser routing performance and should be interpreted alongside the original 77-class scores. They do not independently validate risk detection or complaint-state handling.

To establish whether the transformer representation provided a material improvement, the selected model was compared with a trivial majority baseline and with TF-IDF classifiers using the same train/development/test protocol. Table A4 reports test Macro-F1 for the compared configurations.

Table A4 Model comparison on the BANKING77 test split

Model	Test Macro-F1
Majority class	0.0003
TF-IDF + ComplementNB	0.7934
TF-IDF + LogisticRegression	0.8726
TF-IDF + LinearSVC	0.8940
TF-IDF + calibrated LinearSVC	0.8903
MiniLM + LogisticRegression	0.9307

The selected encoder improved Macro-F1 by 0.0367 over the strongest TF-IDF baseline and by 0.1373 over the TF-IDF ComplementNB baseline. The gain is consistent with the encoder's ability to represent paraphrases and contextual similarity that lexical features do not capture. It is nevertheless a benchmark improvement rather than proof of end-to-end superiority: the TF-IDF alternatives were not tuned for the operational route layer, and the final system's response quality also depends on policy retrieval and rule execution.

Error inspection showed that the weakest fine-grained classes were **transfer_not_received_by_recipient** (F1 0.7952), **balance_not_updated_after_bank_transfer** (0.8000) and **pending_transfer** (0.8169). These labels describe closely related transfer states and were frequently confused with one another. Such errors may have limited operational impact when both labels map to the same route, but they can affect the specificity of a clarification or policy lookup. Most remaining confusions also involved adjacent banking concepts, including card arrival versus delivery estimation and declined transfers versus declined card payments.

A particularly relevant confusion was **card_payment_not_recognised** being predicted as **compromised_card**. This error remains on a high-risk escalation path, which demonstrates why route-level evaluation and the independent risk layer matter: a fine-grained mistake does not automatically produce an unsafe downgrade. Conversely, a correct route prediction does not validate

the entities, the retrieved policy evidence or the final action. Those properties require separate evaluation.

Several limitations qualify the result. BANKING77 contains no native product-recommendation label and provides limited coverage of complaints and fraudulent scenarios, so the recommendation route is governed by Module C and evaluated through scenario tests rather than this benchmark. The dataset contains typed customer-service utterances and does not measure automatic speech recognition errors, speaker variation or noisy voice input. Entity extraction and risk detection are deterministic components evaluated through unit and scenario tests rather than a large independently annotated corpus. Finally, the trained classifier always returns one of its 77 labels and does not output an abstention class; confidence thresholds, clarification and escalation must therefore be validated as part of the complete rules workflow.

6.4 End-to-End System Evaluation

The integrated A-E evaluation checked actions, state, evidence, recommendations and customer-facing responses. Results combine earlier frozen runs and subsequent reconciliations across configurations; they do not represent a rerun of the latest code. Each business case was executed three times. These results are separate from the component tests in Sections 6.1-6.3.

6.4.1 Single-turn End-to-End Results

Population	E2E Appropriate Outcome	Resolved Answer or Recommendation	Safe Non-answer Delivery
Core business (40 × 3)	117/120	33/33	84/87
Complex development (36 × 3)	104/108	14/15	90/93
Complex holdout (12 × 3)	33/36	3/3	30/33
Overall business (88 × 3)	254/264	50/51	204/213

In addition, 39 of 40 core scenarios passed all three repetitions, and all six controlled technical fault-injection checks failed closed. Required evidence was present in 24/24 core answer cases. Recommendation permission was also correct for all permitted cases (30/30) and all denied cases (45/45).

The integrated workflow achieves a high level of correct and safe single-turn delivery across both core and complex-language scenarios. The remaining mismatches show that end-to-end correctness remains stricter than module-level accuracy: a response can be safe while still differing from the reviewed workflow or wording expectation.

6.4.2 Multi-turn State and Safety Results

Two complementary multi-turn test populations were evaluated. The legacy fixture contains 24 earlier workflow sequences with 49 turns and focuses on complaint-state continuity, clarification follow-up and safe workflow routing. The expanded reviewed fixture contains 13 manually reviewed sequences and focuses on bounded recent-turn context, reference resolution, topic switching and stale-context prevention.

For the expanded reviewed fixture, the system passed all 117 evaluated turns and all 13 three-repeat sequences. It achieved 30/30 context-reference resolutions, 9/9 explicit topic-switch cases, 15/15 stale-context prevention cases, 3/3 clarification-closure cases and 3/3 unresolved-reference clarification cases.

For the legacy workflow fixture after the bounded-context repair, 29 of 36 evaluated turns matched the expected workflow outcome, with 4 of 6 sequences passing across all repeats. The bounded-context follow-up subset passed 54/54 turns and 18/18 sequences, with no contextual safety violation. The remaining legacy mismatches were safe action mismatches, mainly returning CLARIFY or ABSTAIN when a different safe workflow action was expected; they are therefore reported as workflow limitations rather than unsafe system behaviour.

The results show that the bounded-context design can resolve reviewed follow-up references without allowing stale or ambiguous context to create an unsupported answer. However, the legacy workflow result also shows that safe routing alone is not sufficient: further refinement is required for full agreement with every expected complaint and service workflow outcome.

6.4.3 LLM Delivery Reliability

LLM-supported delivery was evaluated as a constrained component of the complete workflow, rather than as an independent decision-maker. The independent final-response semantic overlay passed 211 of 224 reviewed ANSWER and CLARIFY responses (94.2%). Four additional responses were unscored because the judge was unavailable and are excluded from the denominator. This assessment checks whether customer-facing wording preserves the approved action, required facts and clarification requirement without adding unsupported claims.

The remaining scored failures were principally clarification-quality issues: the response remained safe but did not always request the most specific missing condition required by the reviewed scenario. Across the raw full-system evaluation, the LLM gateway recorded 147 fallbacks over 964 module events, while end-to-end latency was 4,544 ms at p50 and 8,839 ms at p95. Provider failure, malformed structured output or a timeout triggers deterministic fallback. Provider failure, malformed structured output or a timeout triggers a configured deterministic fallback. All six controlled technical fault-injection checks failed closed; this result is limited to the tested faults and configurations. .

The LLM-assisted configuration achieved the reported workflow and response-quality results, while deterministic rules retained authority over service actions. These results describe the evaluated configuration; without a matched comparison, they do not isolate the contribution of LLM assistance. The observed fallback rate and clarification-quality gaps indicate that the LLM should remain a bounded delivery component and that availability and wording quality require continued monitoring.

6.4.4 Separate Local ASR Smoke Evaluation

ASR was evaluated separately from the text-agent end-to-end workflow because speech recognition produces only an unconfirmed transcript. The user may review and edit this transcript before it becomes a normal text request; consequently, ASR error must not be pooled with text-agent retrieval, rules, safety or final-response metrics.

The local ASR evaluation used whisper.cpp with the fixed multilingual large-v3-turbo-q5_0 model on a deterministic 80-clip public CAiRE/ASCEND smoke set. All selected clips were transcribed successfully.

Audio stratum	Successful clips	Mixed-token error rate
Mandarin-English mixed (60 clips)	60/60	21.12%
Mandarin (10 clips)	10/10	26.36%
English (10 clips)	10/10	31.82%

These rates do not directly represent task failure: minor token or formatting differences may not change the customer's intended request, and the transcript is editable before submission. The downstream workflow therefore processes user-confirmed text rather than raw ASR output.

All selected public clips produced transcripts under the tested configuration. The evaluation did not separately measure recognition accuracy for banking terminology, customer names or financial amounts. These remain priorities for domain-specific testing because recognition errors could alter the interpreted request. The present results therefore demonstrate local transcription on the selected clips rather than banking-domain recognition quality or autonomous voice-banking capability.

7 Findings and Discussion

The findings support the feasibility of NovaBank's modular architecture within the evaluated scenarios. The evidence now extends beyond component performance to integrated decisions, multi-turn behaviour and customer-facing responses. Nevertheless, these evaluation layers establish different capabilities. The fixed mapping from BANKING77 labels to operational routes illustrates the importance of label granularity discussed in classification research **[28]**: some fine-grained errors leave the service route unchanged, but correct routing does not establish complete understanding of the customer's request.

The retrieval results demonstrate why relevant passages must remain subject to evidence checks. Restriction-sensitive and multi-chunk questions were harder than routine supported-service questions because an answer may depend on conditions distributed across passages. Fact-verification research similarly distinguishes finding evidence from determining whether it supports a claim **[29]**. NovaBank's retrieval and lexical checks provide useful safeguards, but cannot independently guarantee the factual correctness of generated responses.

Integrated recommendation checks provide more direct evidence of the permission boundary. The reviewed scenarios correctly permitted eligible invocations and denied prohibited ones, supporting the separation between C's authorisation and D's product selection. This establishes behaviour under configured service constraints rather than financial suitability. The synthetic profiles and demonstration products do not measure customer relevance, satisfaction or financial outcomes.

Multi-turn evaluation revealed both progress and remaining workflow gaps. The expanded context scenarios passed their structured checks, whereas the legacy complaint-state subset retained conservative action mismatches. These findings reinforce the distinction between language

understanding, dialogue-state tracking and response generation **[30]**. The independent semantic assessment further showed that an acceptable **CLARIFY** action can still produce an unhelpful question when its wording does not request the required condition. Structural correctness and conversational usefulness therefore require separate assessment.

Operational results also qualify the system's capabilities. LLM fallback events and observed latency identify reliability and responsiveness as continuing concerns. The separate ASR smoke test demonstrates local transcription on selected public clips, but does not establish banking-domain recognition quality or spoken task completion. Consistent with broader AI-safety concerns [31], unintended behaviour must be assessed alongside task success. Structured traces support investigation of these outcomes; their presence alone does not establish safety or deployment readiness.

8 Conclusion and Future Work

This project developed and evaluated a modular banking-assistant prototype that separates language understanding, policy retrieval, deterministic service control and product recommendation. Its main contribution is an explicit boundary between interpreting customer requests and authorising service actions, supported by structured interfaces and decision traces.

The evaluation provides evidence at both component and integration levels. The intent classifier achieved a BANKING77 test Macro-F1 of **0.9307**, while policy retrieval achieved **95.07% Recall@5** on **71 evidence-bearing development questions**. Across **88 single-turn business scenarios**, each repeated three times, the integrated workflow produced appropriate structured outcomes in **254/264 attempts**. Expanded multi-turn scenarios passed their reviewed structural checks, and independent response assessment passed **211/224 scored responses**. A separate local ASR smoke test demonstrated transcription capability without establishing an integrated voice-banking service.

The results also identify the limits of the prototype. Legacy complaint workflows retained action mismatches, and some clarification responses failed to request the required information. Retrieval relevance does not guarantee factual sufficiency, synthetic recommendations do not establish financial suitability, and public speech clips do not represent banking-domain voice performance. The findings therefore support a controlled prototype rather than deployment readiness or regulatory compliance.

Future work should prioritise the observed complaint-continuity and clarification failures, followed by held-out evaluation of conditional evidence, mixed-intent risk precedence and module failures. Further development should complete authenticated case-resolution workflows, assess recommendation relevance using independent judgements, and evaluate spoken interaction using the implemented transcript-confirmation interface. These steps would extend the demonstrated capabilities while preserving explicit service permissions and human oversight.

9 References

[1] Robertson, S., & Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4), 333–389. https://doi.org/10.1561/1500000019

[2] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 3982–3992. https://doi.org/10.18653/v1/D19-1410

[3]Monetary Authority of Singapore. (2013). *Guidelines on fair dealing - Board and senior management responsibilities for delivering fair dealing outcomes to customers* (Guidelines No. FAA-G11). MAS.

[4] MoneySense. (n.d.). *Resolving a dispute with a financial institution*. Government of Singapore. Retrieved October 2026, from https://www.moneysense.gov.sg/articles/2018/10/resolving-a-dispute-with-a-financial-institution

[5] Geifman, Y., & El-Yaniv, R. (2017). Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS 2017)* (Vol. 30, pp. 4878–4887).

[6] Personal Data Protection Commission Singapore. (2021). *Advisory guidelines on key concepts in the Personal Data Protection Act*. PDPC. https://www.pdpc.gov.sg/guidance/guidelines/advisory-guidelines

[7] DBS Bank. (n.d.). *Meet digibot: Your 24/7 digital banking assistant*. DBS Singapore. Retrieved October 2026, from https://www.dbs.com.sg/personal/support/bank-general-digibot.page

[8] United Overseas Bank. (n.d.). *Contact us & digital assistance: UOB TMRW*. UOB Singapore. Retrieved October 2026, from https://www.uob.com.sg/personal/contact-us.page

[9] OCBC Bank. (n.d.). *Contact us and self-service support*. OCBC Singapore. Retrieved October 2026, from https://www.ocbc.com/personal-banking/contact-us

[10] Revolut Ltd. (n.d.). *Customer artificial intelligence and automated support notice*. Revolut. Retrieved October 2026, from https://www.revolut.com/legal/privacy/

[11] Casanueva, I., Temčinas, T., Gerz, D., Henderson, M., & Vulić, I. (2020). Efficient intent detection with dual sentence encoders. In *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI* (pp. 38–45). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.nlp4convai-1.5

[12] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS 2020)* (Vol. 33, pp. 9459–9474).

[13] Murtaza, S. S., Nie, Y., Avan, E., Soni, U., Liao, W., Carnegie, A., Mathias, C. J., Jiang, J., & Wen, E. (2025). Implementing retrieval augmented generation technique on unstructured and structured data sources in a call center of a large financial institution. In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: Industry Track) (pp. 598–606). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.naacl-industry.48

[14] Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M. (2020). MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained Transformers. In *Advances in Neural Information Processing Systems (NeurIPS 2020)* (Vol. 33, pp. 5776–5788).

[15] Rebedea, T., Dinu, R., Sreedhar, M. N., Parisien, C., & Cohen, J. (2023). NeMo Guardrails: A toolkit for controllable and safe LLM applications with programmable rails. In Proceedings of the

2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (pp. 431–445). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-demo.40

[16] Felfernig, A., & Burke, R. (2008). Constraint-based recommender systems: Technologies and research issues. In Proceedings of the 10th International Conference on Electronic Commerce (ICEC '08) (pp. 1–10). Association for Computing Machinery. https://doi.org/10.1145/1409540.1409544

[17] Larson, S., Mahendran, A., Peper, J. J., Clarke, C., Lee, A., Hill, P., Kummerfeld, J. K., Leach, K., Laurenzano, M. A., Tang, L., & Lasecki, W. S. (2019). An evaluation dataset for intent classification and out-of-scope prediction. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 1311–1316). https://doi.org/10.18653/v1/D19-1131

[18] Chiticariu, L., Li, Y., & Reiss, F. R. (2013). Rule-based information extraction is dead! Long live rule-based information extraction systems! In Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (pp. 827–832). Association for Computational Linguistics. https://aclanthology.org/D13-1079/

[19] Barnett, S., Kurniawan, S., Thudumu, S., Brannelly, Z., & Abdelrazek, M. (2024). Seven failure points when engineering a retrieval augmented generation system. arXiv preprint arXiv:2401.05856. https://doi.org/10.48550/arXiv.2401.05856

[20] Sakata, W., Shibata, T., Tanaka, R., & Kurohashi, S. (2019). FAQ retrieval using query-question similarity and BERT-based query-answer relevance. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 1113–1116). https://doi.org/10.1145/3331184.3331326

[21] Fowler, M. (2002). *Patterns of enterprise application architecture* (pp. 488–495). Addison-Wesley Professional.

[22] Jannach, D., Zanker, M., Felfernig, A., & Friedrich, G. (2010). *Recommender systems: An introduction* (pp. 77–106). Cambridge University Press. https://doi.org/10.1017/CBO9780511763113

[23] Kiros, R., Zhu, Y., Salakhutdinov, R., Zemel, R., Urtasun, R., Torralba, A., & Fidler, S. (2015). Skip-thought vectors. In *Advances in Neural Information Processing Systems (NeurIPS 2015)* (Vol. 28, pp. 3296–3304).

[24] Hendrycks, D., Mazeika, M., & Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In *International Conference on Learning Representations (ICLR 2019)*.

[25] Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020). Beyond accuracy: Behavioral testing of NLP models with CheckList. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 4902–4912). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.442

[26] Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W.-t. (2020). Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 6769–6781). https://doi.org/10.18653/v1/2020.emnlp-main.550

[27] Sokolova, M., & Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4), 427–437. https://doi.org/10.1016/j.ipm.2009.03.002

[28] Silla, C. N., & Freitas, A. A. (2011). A survey of hierarchical classification across different application domains. *Data Mining and Knowledge Discovery*, 22(1–2), 31–72. https://doi.org/10.1007/s10618-010-0175-9

[29] Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: A large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)* (pp. 809–819). https://doi.org/10.18653/v1/N18-1074

[30] Rastogi, A., Zang, X., Sunkara, S., Gupta, R., & Khaitan, P. (2020). Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05), 8689–8696. https://doi.org/10.1609/aaai.v34i05.6394

[31] Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*. https://doi.org/10.48550/arXiv.1606.06565

10 Appendix

10.1 Declaration

We used Generative AI tools (including ChatGPT and Gemini) to assist in brainstorming the report outline, structuring sections, refining language, and improving expressions throughout this report.

All core ideas, system design architectures, reasoning workflows, and evaluations were conceptualized, reviewed, and finalized by the authors. We take full responsibility for the originality, content, and quality of the submitted work.

10.2 Mapped System Functionalities against Modular Courses

Courses	**Knowledge and Techniques**
Machine Reasoning (MR)	• Rule-Based Reasoning • Structured Knowledge Representation • Constraint-Based Eligibility Checking • Deterministic Decision Rules and Prioritisation
Reasoning Systems (RS)	• Hybrid Reasoning Architecture • Retrieval-Augmented Generation (RAG) • Multi-Signal BM25 Retrieval • Transformer-Based Cross-Encoder Reranking • Knowledge-Based Product Recommendation
Cognitive Systems (CGS)	• Natural Language Understanding and Intent Recognition

	• Dialogue Context and State Management • Uncertainty Handling: CLARIFY and ABSTAIN • Evidence-Grounded Response Design

10.3 Installation and User Guide

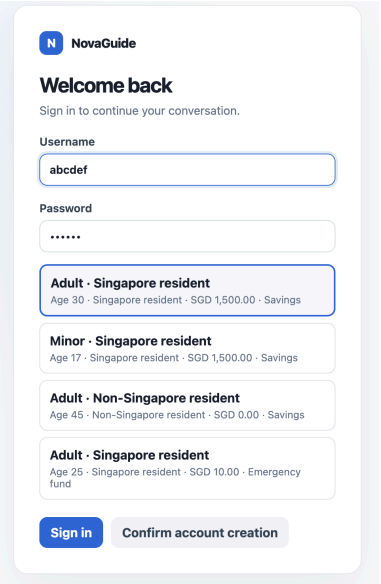
NovaBank is delivered as a local demonstration prototype using synthetic data only. It requires a Python environment, a modern browser and the required local model files. The detailed installation commands, dependency versions, model download instructions and environment-variable examples are maintained in the project GitHub repository.

The normal text workflow uses a frozen all-MiniLM-L6-v2 sentence encoder for intent understanding and an ms-marco-MiniLM-L6-v2 Cross-Encoder for retrieval reranking. These models must be downloaded and available locally before the production composition is started. If a required model is unavailable, the application fails safely rather than silently substituting another model.

Optional voice input requires a local whisper.cpp installation and the pinned multilingual Whisper large-v3-turbo-q5_0 model. The component converts a short recording into an editable transcript. Audio and unconfirmed transcripts are not stored, and only text confirmed by the user enters the normal agent workflow. Voice input is speech-to-text only; text-to-speech is not implemented.

Optional LLM-assisted delivery requires a server-side OpenAI-compatible provider configuration, including a provider endpoint, API credential and supported model name. These values are held in local server configuration and are never exposed to the browser. The frozen delivery profile enables guarded LLM support for understanding, query rewriting and response wording, while the deterministic RuleEngine remains responsible for actions, state transitions, evidence permission and recommendation permission. Module D determines product eligibility and ranking. Full provider configuration and reproducibility steps are available in the GitHub README.

To use the web application, the evaluator starts the local server according to the repository instructions and opens the local browser address. The evaluator creates a local account, selects a synthetic demo profile, signs in and starts a new conversation.



A request can then be entered in the chat interface. Depending on the rules outcome and available evidence, the system may provide an evidence-grounded answer, ask for clarification, abstain, present escalation or complaint guidance, or show eligible recommendations. Evidence-backed answers provide supporting policy excerpts and citations through the interface.

When local ASR is enabled, the evaluator may select the Voice button, record a short request, review or edit the returned transcript and select Use text before sending it as a normal message. The prototype does not access real banking accounts, execute transactions or make binding financial decisions.

10.4 Project Proposal:

Project Proposal Document Title: Banking Voice Agent for Complaint Resolution and Responsible Next Best Action Recommendations

Date Approved: 13 September 2026 Original

Document Link: [Shaojunzhe_Group4_Banking_Voice_Agent_Proposal]

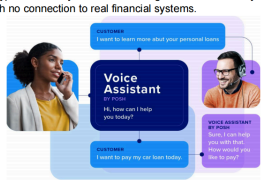
—---------- project proposal—----------

Banking Voice Agent for Complaint Resolution and Responsible Next Best Action Recommendations

Course: NUS ISS Certificate in Intelligent Reasoning Systems
Group Number: Group [34]
Group Members: Shao Junzhe(A0356163A) Lin Zhi(A0356103N), Ruan Yu (A0351861A) , Zhao Xinyu(A0352367Y), Xie Yuchen(A0350890B)
Date: 13 September 2026

1. Introduction

This runnable, voice-enabled retail banking prototype understands customer speech, identifies intent and risk, retrieves relevant banking knowledge, supports complaint resolution, and recommends appropriate next best actions. By combining machine learning with deterministic business rules, it balances conversational flexibility with operational safety. Unlike a general-purpose chatbot, high-risk service actions and escalations are governed by explicit rules and confidence thresholds rather than unconstrained LLM generation. For transparency, the interface displays retrieved evidence, routing decisions, confidence information, and policy checks. The prototype uses only fictional banking scenarios and synthetic customer and transaction data, with no connection to real financial systems.



Banking customer service presents a fundamental challenge: customers expect natural and flexible conversations, while banks require consistent procedures, privacy protection, and clear safety controls. Traditional rule-based systems can be too rigid for open-ended requests, whereas unconstrained generative AI may produce unsupported information or inappropriate actions in sensitive situations. This project addresses that gap through a hybrid architecture that combines AI-based language understanding and retrieval with deterministic business rules. Generative AI supports natural communication, while explicit rules govern complaint handling, eligibility checks, escalation, and product recommendations. The prototype demonstrates how intelligent reasoning can improve service automation while keeping decisions transparent, auditable, and safely constrained.

Project Goals

- **Build an End-to-End Voice Banking Prototype:** Develop an English voice interface for a synthetic Singapore retail bank, with English–Mandarin support as an optional extension.
- **Detect Customer Intent and Risk:** Classify banking intents and map requests to appropriate service queues, severity levels, and permitted actions with confidence-based abstention.
- **Provide Grounded Banking Answers:** Answer policy and product enquiries using a curated knowledge base with source citations, while rejecting unsupported responses.[1]
- **Implement Safe Complaint Handling:** Resolve or escalate complaints through auditable business rules, with priority handling for fraud, lost cards, and unauthorised transactions.[2]
- **Recommend Responsible Next Best Actions:** Suggest products or service actions only when complaint status, customer consent, eligibility, and risk rules allow.[3]
- **Evaluate System Performance:** Compare major modules against suitable baselines using quantitative metrics and present the results through the system interface and final report.

2. Project Background and Market Context

- **Contextual Background**:
  The project operates within the retail banking customer service domain, specifically focusing on voice-based interactions. In this environment, service calls frequently mix simple administrative inquiries with emotionally expressed complaints and urgent fraud concerns. Furthermore, operating within the Singapore regulatory context requires the system to align with structured complaint and escalation frameworks outlined by the Association of Banks in Singapore (ABS) and MoneySense, alongside strict data privacy obligations regarding voice recordings under the PDPC. [4]

- **Problem Statement & Key Challenges:**
  The core bottleneck is balancing conversational flexibility with mandatory financial compliance and risk management. Traditional, rule-based IVR systems fail because they cannot capture free-form intent, retain multi-turn context, or ground explanations in complex policy documents. Conversely, naive generative AI methods are dangerous in this domain; unconstrained LLMs may hallucinate unsupported answers, leak account-specific data prior to authentication, or inappropriately offer marketing products during sensitive fraud complaints.

- **Landscape Overview:**
  The current market is heavily investing in automated voice self-service and conversational banking workflows. Existing industry approaches generally fall into two extremes: rigid, rule-based IVR systems that lack natural conversational understanding, or black-box generative AI platforms that struggle to guarantee deterministic compliance. While the industry is actively shifting towards AI-driven guided resolution and automated human handoff, there remains a critical market gap for a highly transparent, auditable framework. The industry currently lacks solutions that explicitly isolate and expose their reasoning components to ensure that high-risk financial escalations are never overridden by generative models.

3. Market Research

- **Related research and technical foundation:**
  This project is mainly based on three types of research: bank intent identification, RAG, and hybrid reasoning. Casanueva et al. (2020) proposed the BANKING77 dataset in "Efficient Intent Detection with Dual Sentence Encoders", which contains 13,083 bank customer queries and 77 types of fine-grained intents, providing a mature benchmark for bank customer service intent identification [5]. This project will conduct intent classification and business diversion based on this dataset.

  In terms of knowledge questioning, Murtaza et al. (2025) published a study in the NAACL Industry Track that deployed RAG in large financial institutions' Call Centers [6]. It improved the factual reliability of answers by using external unstructured and structured data, and focused on low latency and high reliability. Based on this, this project further combines RAG with business status, risk rules, and manual transfer: LLM is responsible for language understanding and generation, while high-risk decisions such as fraud, complaint status, marketing permission, and product eligibility are controlled by deterministic rules. This is also consistent with the current design of the project, which states that LLM is responsible for understanding and expression, while high-risk decisions are handled by state machines and rules.

- **Market & Competitor Analysis:**
  Bank customer service is gradually evolving from traditional IVR and rule-based robots to Conversational AI and generative AI-assisted customer service. The current market-related solutions can be roughly divided into two categories.

  The first category is the bank's own virtual assistants. Erica of Bank of America can help users query account information, search for transactions, manage bank cards, view spending situations, and transfer to human customer service when unable to answer questions [7]. It is worth noting that Bank of America officially stated that Erica is currently mainly based on natural language processing and predefined responses, rather than directly using generative large models. This reflects the bank's emphasis on controllability in the scenario. Eno from Capital One is also a bank-owned intelligent assistant [8]. It mainly offers account-related inquiries, balance checks, consumption reminders, suspicious transaction alerts, and credit card services. It supports multiple interaction channels such as apps, web pages, and text messages.

  The second type is the Contact Center AI platform for enterprises. Google Cloud's Contact Center AI Platform (CCAI) offers capabilities such as multi-channel customer interaction, AI virtual customer service, intelligent routing, Agent Assist, and manual customer service transfer [9]. The virtual customer service can serve as the first line of support and automatically upgrade to manual customer service when there is insufficient knowledge or abnormal situations occur.

  Genesys Cloud CX is another mature enterprise-level contact center platform that offers capabilities such as multi-channel interaction, intelligent routing, virtual agents, customer service copilot, customer journey management, and workflow integration [10]. The key feature lies in coordinating human customer service with AI agents to complete the end-to-end customer service process. PolyAI, on the other hand, focuses more on voice AI agents, supporting typical customer service tasks such as account management, authentication, billing and payments, FAQs, and call routing [11].

- **User Needs & Industry Demands:**
  The intelligent customer service of banks mainly addresses three types of demands:
  Bank customers. The customers expect to obtain answers to their questions quickly without going through complicated phone menus or waiting for long periods for human customer service. At the same time, for sensitive issues such as fees, complaints, and suspicious transactions, the customers need to know the basis for the system's responses and hope to be smoothly transferred to human assistance when the AI cannot handle it reliably, rather than constantly communicating with the robot repeatedly.

  Bank customer service staff. The human customer service staff need to handle a large number of repetitive inquiries and frequently query policies and the background of customer questions. When the AI transfers the question to human assistance, if it can simultaneously provide the customer's intention, risk assessment, historical conversations, policies that have been completed, and the processing steps that have been done, it can reduce the cost of human re-understanding of the question.

  Bank operations and compliance personnel. For banks, the level of automation is not the only goal. The system must also be able to explain why an operation is performed, which policies are used, which rules are triggered, and why a human transfer was made. Especially in scenarios such as complaints, fraud, and product recommendations, the system needs to avoid directly converting unverified large model outputs into business decisions.

Overall, the existing products have demonstrated the commercial feasibility of voice interaction, natural language understanding, knowledge questioning, intelligent routing, and human transfer in customer service. Therefore, this project does not consider "building bank customer service using LLM" itself as the main innovation, but rather focuses more on controlled automation in the banking scenario.

Compared to general customer service scenarios, complaints, suspicious transactions, unauthorized transactions, customer vulnerability, and financial product recommendations in the banking business all have higher risks. Therefore, the system not only needs to answer customer questions, but also needs to determine: whether the current issue is suitable for automatic processing, when it must be transferred to human, and whether product

recommendations can be made at the moment.

Therefore, this project adopts a hybrid approach of language understanding,explicit business state and deterministic rules, enabling the system not only to decide "how to answer", but also to decide whether to continue automatic processing. This is consistent with the existing design of the project: for fraud, unauthorized transactions, or low-confidence scenarios, priority is given to transferring to human, and product recommendations can only be made after the issue has been resolved, the customer has explicitly agreed, and the qualification conditions are met.

4. Project Scope & Constraints

The project will focus on a defined set of service scenarios for a fictional Singapore retail bank: routine enquiries, fee complaints, suspected unauthorised transactions, and next best action recommendations after a service issue has been resolved. English speech input and spoken responses will form the core interaction, while English–Mandarin support will remain an optional extension. Machine learning will support intent classification, and risk assessment will identify requests that require clarification or priority escalation. Retrieval will provide relevant policy passages for the language model to explain with source citations. Deterministic rules will govern complaint-state transitions, permitted actions and recommendation conditions. Product recommendations will be considered only after the service issue has been resolved and customer consent, product eligibility, risk and confidence checks permit them. Creating a complaint case will not, by itself, count as resolving the complaint.

The prototype will use synthetic customer and transaction records, with simulated account operations, case handling and human handoff. It will not connect to real financial systems, authenticate real customers, execute financial transactions or make binding financial decisions. These boundaries allow the group to investigate the reasoning workflow without requiring access to banking infrastructure or customer records. Model and service selection will depend on available computing resources and access conditions, and response latency will be measured during testing. Speech recognition errors, incomplete retrieval and incorrectly specified rules may still affect system behaviour. Evaluation will therefore examine performance and failure cases within the selected scenarios; it will not establish regulatory compliance or readiness for deployment in a real bank.

Area	Core scope	Boundary or extension
Voice interaction	English speech input and spoken responses	English–Mandarin support is optional
Intent and risk assessment	Identify service needs and risk signals to support routing	Uncertain or high-risk requests require clarification or escalation
Knowledge-based answers	Retrieve policy passages and provide answers with citations	Unsupported questions require clarification or handoff
Complaint handling	Track complaint status and simulate case creation and escalation	Creating a case does not mean that the complaint is resolved
Next best action	Offer recommendations after resolution, subject to consent, eligibility and risk checks	Product recommendations are blocked while a complaint remains unresolved
Banking operations	Use synthetic customer and transaction records and simulated tools	No real banking connections or financial transactions

5. Data Collection and Preparation

The proposed reasoning system combines public datasets, public or curated domain knowledge, and simulated data to build its foundational knowledge and support different components of the banking voice customer service agent. No real customer accounts or real banking transactions are required for the prototype; instead, synthetic customer profiles, transactions, complaints, eligibility conditions, consent states, and high-risk scenarios are used to support safe system development and evaluation. This is consistent with the project scope, which focuses on a fictional Singapore retail bank rather than deployment in a real banking environment.

Sources of Data / Knowledge. Several public datasets are used for different learning tasks. BANKING77 provides labelled banking queries across 77 fine-grained intent categories and serves as the primary source for training and evaluating the intent classification component. These intents will be mapped into the smaller set of operational routes required by the proposed customer service workflow. CFPB Consumer Complaint Data can be used to analyse complaint products, issue categories, and complaint outcomes, although its U.S.-specific context must be considered when adapting insights to the fictional Singapore banking scenario. UCI Bank Marketing provides data for developing a small marketing-response or recommendation baseline, with leakage-prone attributes such as call duration removed. Santander product recommendation data may optionally support experiments on multi-product recommendation, using appropriate sampling and temporal splitting to reduce leakage.

In addition to structured datasets, the system uses bank policies, FAQs, and relevant public guidance as its domain knowledge source for Retrieval-Augmented Generation (RAG). Documents are maintained with metadata such as source, section, date, and version so that generated responses can be traced back to supporting evidence. Synthetic cases complement these sources by representing customer profiles, transaction histories, complaint situations, product eligibility, consent status, and safety-critical scenarios that may be rare or unsuitable to obtain from real customer data.

Acquisition and Ingestion. Raw datasets are first collected from their respective public sources and converted into a consistent internal format. The preprocessing pipeline includes removing duplicates and unusable records, standardising category labels and field names, handling missing values, removing leakage-prone attributes, and mapping dataset-specific labels into the project's operational taxonomy. For example, the 77 BANKING77 intents are mapped into customer-service routes such as ordinary enquiries, complaints, and risk-sensitive cases. Structured customer and product attributes are similarly normalised so that they can be consumed by the rule and recommendation components.

For the RAG knowledge base, policy documents and FAQs are cleaned and divided into semantically meaningful text chunks. Each chunk is associated with metadata such as document source, section, effective date, and version, after which it is indexed for both keyword and vector retrieval. The project specifically adopts a hybrid retrieval approach combining keyword and vector search, with responses required to provide supporting sources. This ingestion pipeline therefore transforms heterogeneous raw information into auditable inputs that can be used consistently by the classifier, retrieval system, rule engine, and recommendation module.

Knowledge Engineering and Representation. Different forms of knowledge are represented according to their role in the reasoning process. Unstructured banking policies and FAQs are represented as text chunks and vector embeddings for semantic retrieval, together with keyword indexes and document metadata for hybrid RAG. Structured operational knowledge—including identity verification requirements, complaint states, fraud and safety conditions, marketing consent, product eligibility, and escalation criteria—is encoded as deterministic rules and state transitions rather than being left entirely to the language model. This distinction is important because the project assigns language understanding and natural-language generation to the model while reserving high-risk decisions for the state machine and deterministic rule engine.

Customer and product information is represented through structured feature records, while recommendation knowledge is organised as a two-stage decision process: hard constraints first filter out ineligible or unsafe products, and a ranking model then orders the remaining candidates. This representation supports the project's requirement that recommendations should only occur after the original issue has been resolved and the customer has provided the required consent. The overall system therefore uses a hybrid knowledge representation consisting of embeddings for semantic knowledge, structured metadata for provenance, deterministic rule tables/state machines for safety-critical reasoning, and learned representations for classification and recommendation.

Data Challenges and Mitigation. Several data limitations are expected. First, public banking datasets may not perfectly match the target setting. In particular, CFPB complaint data reflects a U.S. regulatory and consumer context, while the prototype represents a fictional Singapore retail bank. Domain mismatch is mitigated by treating such datasets primarily as sources for complaint patterns and experimental data rather than as authoritative policy knowledge; actual system answers are grounded in the curated policy and public-guidance knowledge base.

Second, data scarcity and class imbalance may occur for safety-critical but relatively rare scenarios such as fraud, unauthorised transactions, vulnerable customers, or severe complaints. Synthetic cases can be deliberately generated to increase coverage of these scenarios and create targeted safety test sets. Third, speech input introduces noise through ASR errors, accents, background noise, and incorrectly recognised banking entities. The system therefore uses confidence thresholds and confirmation steps, particularly for sensitive entities such as account information, monetary amounts, and dates.

Finally, noisy labels, sparse recommendation histories, and limited manually labelled examples may restrict supervised learning. These problems are mitigated through label mapping, data validation, careful train/test or temporal splits, synthetic scenario generation, and the use of simple baselines before introducing more complex models. For RAG, unsupported or low-confidence retrieval is handled conservatively: if no reliable source can be retrieved, the system should abstain rather than fabricate an answer and escalate the case to a human agent. The retrieved evidence and policy version are also recorded for auditability.

Overall, the reasoning system does not rely on a single knowledge representation or dataset. It combines public labelled datasets for learning, curated policy documents for grounded knowledge, synthetic data for controlled and safety-critical scenarios, vector representations for semantic retrieval, and deterministic rules for high-risk decisions. This hybrid design supports the project's broader objective of building an auditable customer-service system with explicit business workflows and safety boundaries rather than an unrestricted conversational chatbot.

6. System Design & Intelligent Reasoning Architecture

System Architecture.
The system adopts a modular pipeline: customer speech is converted into text by ASR, followed by intent classification, risk detection, and entity extraction. A state machine and deterministic rule engine then decide whether to retrieve policy knowledge through RAG, handle a complaint, provide a compliant recommendation, or escalate the case to a human agent. The final response is delivered through text and TTS with supporting citations and decision traces.

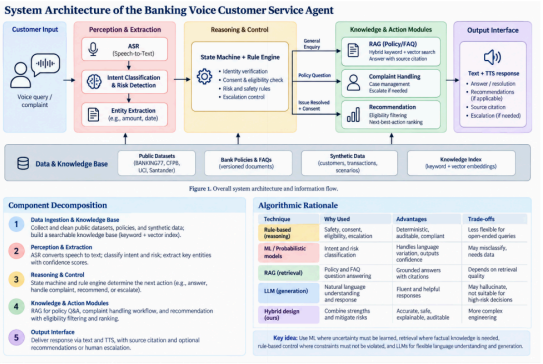


Figure 1. Overall system architecture and information flow.

Component Decomposition.
The system consists of five main modules: (1) Data Ingestion & Knowledge Base, which processes public datasets, bank policies, FAQs, and synthetic cases; (2) Perception & Extraction, which performs ASR, intent/risk classification, and entity extraction; (3) Reasoning & Control, where the state machine and rules enforce safety, consent, eligibility, and escalation requirements; (4) Knowledge & Action, where hybrid RAG retrieves policy information and the recommendation module ranks eligible next-best actions; and (5) Output Interface, which provides text/TTS responses, citations, recommendations, and escalation results.

Algorithmic Rationale.
A hybrid reasoning approach is selected instead of relying on a single technique. Machine-learning models are suitable for uncertain tasks such as intent and risk classification, while rule-based reasoning is preferred for safety-critical decisions because it is deterministic and auditable. Hybrid RAG combines keyword and vector retrieval to provide grounded policy answers with citations. The LLM is mainly responsible for language understanding and response generation rather than high-risk decision-making. This design balances the flexibility of ML/LLMs with the safety, explainability, and compliance of deterministic rules.

Identified Risk / Bottleneck	Impact Level (H/M/L)	Mitigation Strategy & Contingency
Reasoning latency / Computational overhead	Medium	Pruning search spaces, caching intermediate inference states.
Data quality / SME availability	High	Synthetic data augmentation and structured interview templates.
Model hallucination / logical inconsistency	High	Hybrid neuro-symbolic verification layer and deterministic rule checks.

9. Conclusion

We propose a banking voice agent that combines intent and risk identification with evidence-based answers, rule-governed complaint handling and conditional next best action recommendations. The intended contribution is a working prototype that makes its decisions and supporting evidence visible: users will be able to see why a request follows a particular service path, which policy passages support an answer, and why a recommendation is offered or withheld. This design provides a concrete setting for applying intelligent reasoning to service tasks that require both natural communication and explicit procedural controls.

We will evaluate the proposed methods against suitable baselines and test whether the workflow follows its defined rules, including escalation conditions and the blocking of product recommendations during unresolved complaints. The evaluation will distinguish the quality of model outputs from the correctness of workflow execution and will document failures as well as successful cases. The resulting prototype and findings should help assess the usefulness and limitations of this approach within the simulated banking setting, without assuming that performance on public benchmarks or synthetic cases will transfer directly to real customer service.

10. Supplementary Materials

- **Appendix A: AI Tool Usage Declaration:**
  We used Generative AI tools (including ChatGPT and Gemini) to assist in brainstorming the report outline, structuring sections, refining language, and improving expressions throughout this proposal.

7. Preliminary Results & Evaluation Metrics

- **Quantitative & Qualitative Metrics:**

Metric	Evaluate purpose	Measure Methodology
Intent Macro-F1	Accuracy of banking intent recognition	Evaluate predicted intent labels against ground-truth labels on the BANKING77 test set and calculate Macro-F1 across all intent classes.
RAG Recall@5	Ability to retrieve the correct banking policy or knowledge	Construct a Question–Gold Policy test set and calculate the percentage of queries for which the correct document appears in the top 5 retrieved results.
Safe Escalation Rate	Ability to correctly identify and escalate high-risk cases	Use predefined high-risk scenarios, such as suspected fraud and unauthorized transactions, and calculate the proportion that are correctly identified and transferred to human agents.
Recommendation Violation Rate	Compliance and safety of product recommendations	Measure the percentage of recommendations that violate eligibility, customer consent, unresolved complaint, or other predefined safety rules.
End-to-End Task Completion Rate	Overall effectiveness of the complete customer-service workflow	Define success criteria for each test scenario and calculate the percentage of cases in which the system completes all required steps correctly.

8. Challenges, Risks & Roadblocks

All core ideas, system design architectures, reasoning workflows, and evaluations were conceptualized, reviewed, and finalized by the authors. We take full responsibility for the originality, content, and quality of the submitted work.

- **Appendix B: References:**

[1] A. Asai, Z. Wu, S. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection," in *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[2] T. Rebedea et al., "NeMo Guardrails: A Toolkit for Controllable and Safe LLM Applications with Programmable Rails," *arXiv preprint arXiv:2310.10501*, 2023.

[3] Q. Wu et al., "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[4] Monetary Authority of Singapore (MAS), "Principles to Promote Fairness, Ethics, Accountability and Transparency (FEAT) in the Use of Artificial Intelligence and Data Analytics in Singapore's Financial Sector," MAS, Singapore, 2018. [Online]. Available: https://www.mas.gov.sg.

[5] I. Casanueva, T. Temčinas, D. Gerz, M. Henderson, and I. Vulić, "Efficient Intent Detection with Dual Sentence Encoders," in *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, pp. 38–45, 2020, doi: 10.18653/v1/2020.nlp4convai-1.5.

[6] S. S. Murtaza, Y. Nie, E. Avan, U. Soni, W. Liao, A. Carnegie, C. J. Mathias, J. Jiang, and E. Wen, "Implementing Retrieval Augmented Generation Technique on Unstructured and Structured Data Sources in a Call Center of a Large Financial Institution," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 3: Industry Track*, pp. 598–606, 2025, doi: 10.18653/v1/2025.naacl-industry.48.

[7] Bank of America, "Erica® – Virtual Financial Assistant," *Bank of America*. [Online]. Available: Bank of America Erica. [Accessed: Sep. 12, 2026].

[8] Capital One, "Eno, your Capital One assistant," *Capital One*. [Online]. Available: Capital One Eno. [Accessed: Sep. 12, 2026].

[9] Google Cloud, "CCaaS (CCAI Platform)," *Google Cloud Documentation*. [Online]. Available: Google Cloud Contact Center AI Platform. [Accessed: Sep. 12, 2026].

[10] Genesys, "Genesys Cloud CX – AI-Powered Experience Orchestration Platform," *Genesys*. [Online]. Available: Genesys Cloud CX. [Accessed: Sep. 12, 2026].

[11] PolyAI, "Voice AI Agents for Customer Service," *PolyAI*. [Online]. Available: PolyAI. [Accessed: Sep. 12, 2026].